Corrective Overoptimism

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Abstract

Booms in infrastructure investment – railways in the 1840s, electricity in the 1920s, fibre in the 1990s – often coincide with investor exuberance and losses. But the infrastructure remains long after these episodes. I show that investor overoptimism can serve as a second-best corrective force against chronic underinvestment: when private promoters cannot appropriate the network, knowledge, and allocative spillovers of connective infrastructure, diagnostic expectations over-extrapolate good news and induce irreversible investments, pushing capital accumulation toward the socially efficient benchmark. This mechanism – corrective overoptimism – is embedded in a model of spatial development with endogenous infrastructure and estimated on data from England and Wales, 1700–1881, spanning the canal and railway booms. Corrective overoptimism can be a potent channel of growth.

JEL Classification: D84, E22, E32, N73, O18, O33, R12, R42.

Keywords: Corrective overoptimism, diagnostic expectations, infrastructure investment, spatial development, technology diffusion, Industrial Revolution.

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> Wherever you are in the country, you are just a short distance from the great arterial system. At any moment, you could integrate with this great life-force traversing England; become an essential part of the nation or retreat to the suburban domesticity that you call home.
>
> — Simon Foxell, *Mapping England*

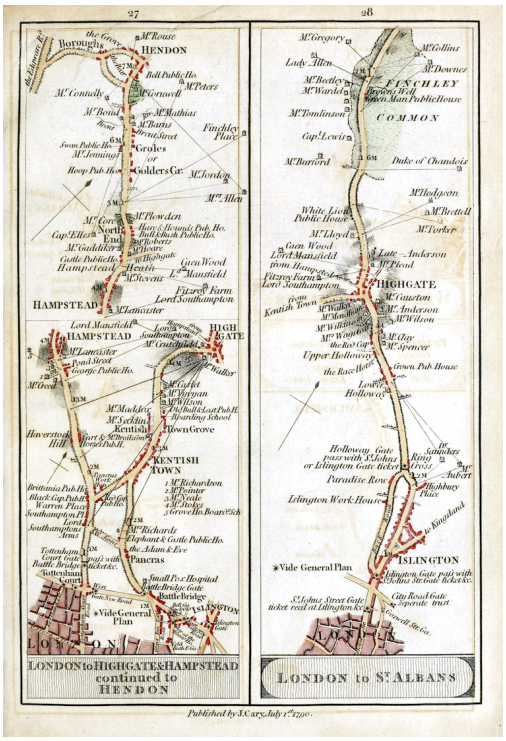
The mass expansion of railways, the electric grid, and the fiber-optic network each coincided with dramatic boom–bust cycles: the British Railway Mania of the 1840s, the electrification boom of the 1920s, and the telecommunications buildout of the late 1990s.[1] Investors in these episodes frequently suffered severe financial losses as initial revenue projections proved overly optimistic. Yet when the financial exuberance subsided, the physical networks remained in place, permanently lowering the cost of moving goods and ideas across space. Janeway (2018) and Morck (2022) argue that this recurrent pattern is not incidental: speculative infrastructure booms have repeatedly laid the physical foundations for modern economic growth. This paper formalizes that hypothesis in a dynamic spatial general equilibrium model and measures its quantitative importance during the British Industrial Revolution.

The mechanism unites endogenous growth and spatial agglomeration with belief formation through the theory of the second best (Lipsey and Lancaster, 1956). Since Lucas (1988), Romer (1990), Krugman (1991), Aghion and Howitt (1992), and Bresnahan and Trajtenberg (1995), a foundational insight of growth and spatial economics is that decentralized markets chronically underprovide durable, general-purpose infrastructure and knowledge. When private promoters sink capital into connective networks before demand is realized, they cannot appropriate inframarginal transit savings across the wider network (Donaldson and Hornbeck, 2016; Fogel, 1964), reallocation gains across distorted production sectors (Hornbeck and Rotemberg, 2024; Hsieh and Klenow, 2009; Restuccia and Rogerson, 2008), or the spatial diffusion of technological discoveries (Desmet and Rossi-Hansberg, 2014; Nelson and Phelps, 1966). Under rational expectations, this appropriation wedge traps decentralized provision below the socially efficient benchmark. If, however, promoters over-extrapolate favorable news about fundamentals—exhibiting the speculative exuberance emphasized by Keynes (1936), Kindleberger (1978), and Shiller (2000)—their heightened willingness to build pushes capital accumulation toward the social optimum. Because installed route capacity is physically irreversible (Arrow, 1968; Dixit and Pindyck, 1994), the networks built during boom waves persist long after beliefs revert. I call this second-best mechanism *corrective overoptimism*.

Whereas a macro-finance tradition studies how rational asset-price bubbles can crowd in investment by relaxing borrowing constraints (Guerron-Quintana et al., 2023; Hirano and Yanagawa, 2016; Martin and Ventura, 2012; Miao and Wang, 2014), my central theoretical contribution is to show how belief overreaction directly counteracts real spillover and appropriation wedges even without financial frictions. Three primitive forces drive the mechanism: (i) a positive wedge between the social and private marginal returns to infrastructure from uninternalized network, allocative, and knowledge spillovers; (ii) forward-looking promoters whose beliefs overreact to fundamental news that expands that wedge; and (iii) physical irreversibility of the installed network.

To discipline belief distortion without ad hoc wedges, I model expectations using diagnostic expectations (DE), developed by Bordalo et al. (2018) as a microfoundation for Kahneman and Tversky (1972)'s representativeness heuristic. Under diagnostic expectations, agents overweight future states whose objective likelihood has increased most in light of recent news. While a growing

[1] See, for example, White (1990), Pástor and Veronesi (2009a), Odlyzko (2010), Greenstein (2015), and Caoui and Steck (2025).



Great North Road out of London, 1790, John Cary. Courtesy of Duke University via HathiTrust.

literature shows that diagnostic expectations account for survey forecast overreaction, credit cycles, and macroeconomic volatility (Bianchi et al., 2024; Bordalo et al., 2026, 2019, 2020; Gennaioli et al., 2016; L'Huillier et al., 2023; Maxted, 2023), existing work largely evaluates overreaction relative to an undistorted or financially constrained benchmark where belief volatility reduces welfare.[2] Here, by contrast, when a diagnostic infrastructure company observes favorable news about manufacturing or transport productivity, it overestimates future local land rents and builds route capacity beyond the rational Markov perfect benchmark. Because installed route capacity is durable, the resulting network expansion permanently accelerates spatial trade integration and technological catch-up.

Section I illustrates this logic in a two-period model of a monopoly export railway with sunk capacity, showing how news-amplified investment raises expected physical welfare when private returns fall short of social returns. Section II documents three historical regularities from England and Wales over 1710–1881 that motivate and discipline the spatial model: (i) a shift of manufacturing toward northern coalfields alongside a nationwide expansion of distribution and transport employment (Bogart et al., 2022; Keibek, 2017; Shaw-Taylor and Wrigley, 2014); (ii) the statutory separation between locally promoted turnpike, canal, and railway companies—which built durable routes to capture local commercial rents—and competitive carriers moving freight over the network (Bogart, 2014; Trew, 2020; Ward, 1974); and (iii) recurrent waves of promoter optimism and overbuilding during the canal and railway manias (Arnold and McCartney, 2011; Campbell, 2012; Odlyzko, 2010). Using Jordà (2005) local projections, shift-share instruments interacting predetermined coalfield and 1710 manufacturing exposure with leave-one-out regional construction cycles,

[2] In complementary work, Flynn and Sastry (2025) and Flynn et al. (2026) study contagious narratives and persistence learning over the business cycle, while Bizzarri and d'Arienzo (2024) analyze the social value of information overreaction in abstract coordination settings.

and railway operating accounts, I show that capacity surges load on current and lagged output cycles, reverse below trend over the next one to three years, and compress subsequent operating yields.

Section III embeds these features into a dynamic spatial general equilibrium model on a continuum of locations partitioned into J counties, building on Rossi-Hansberg (2005), Desmet and Rossi-Hansberg (2014), and Trew (2020). Workers with Stone–Geary preferences (Fajgelbaum and Redding, 2022; Kongsamut et al., 2001) choose residence before private discoveries at fixed contract wages and hold a diversified national portfolio. Agricultural and manufacturing entrants bid for pure site leases, committing production labor and goods-financed innovation expenditures to draw Pareto productivity gains (Desmet and Rossi-Hansberg, 2014; Kortum, 1997) subject to a local manufacturing operating wedge (Desmet et al., 2020; Hornbeck and Rotemberg, 2024). Competitive carriers combine local labor with inherited route capacity to supply county distribution and haulage services, while route infrastructure reduces physical transit losses and accelerates the spatial adoption of frontier technologies (Berkes et al., 2026; Cai et al., 2022; Nelson and Phelps, 1966). Parliamentary-authorized county infrastructure companies play a dynamic Markov perfect investment game under convex construction costs (Hayashi, 1982) and diagnostic expectations.

The model's tractability rests on a clean separation between within-period spatial equilibrium and forward-looking infrastructure investment (Allen and Arkolakis, 2014; Desmet and Rossi-Hansberg, 2014; Fajgelbaum and Schaal, 2020; Kleinman et al., 2023). Annual worker mobility and diversified portfolio ownership reduce residential choice to static indirect-utility equalization (Proposition 1); firm innovation reduces to a reservation-profitability threshold, leaving the within-period equilibrium invariant to diagnosticity θ conditional on county investment (Proposition 2); installed route capacity accelerates spatial knowledge catch-up in closed form (Proposition 3); and county investment satisfies a Tobin marginal-q condition with a unique stage Nash equilibrium under own-cost diagonal dominance (Proposition 4). Walras's law holds across all spatial ledgers (Proposition 5), and Propositions 6 and 7 establish existence and uniqueness of the conditional dated equilibrium and the dynamic Markov perfect equilibrium.

Section IV takes the model to the 1710–1881 trajectories of aggregate and sectoral output, relative industrial prices, occupational shares, spatial sorting covariances, and multimode transport capacity. The Simulated Method of Moments estimates the diagnosticity parameter at $\widehat{\theta} = 0.739$ (SE $= 0.090$), rejecting the rational expectations restriction ($\theta = 0$, $t = 8.21$, $p < 0.0001$) and matching the $\theta \approx 0.74$–0.91 range documented for credit spreads, earnings forecasts, and firm investment (Bordalo et al., 2018, 2026, 2019). When the rational expectations benchmark ($\theta = 0$) is re-estimated, it matches the secular trends in output and employment, but it cannot simultaneously generate the 1830–1850 railway-mania expansion, the post-1850 deceleration, and the negative serial correlation and spatial reversals of infrastructure investment. Out of sample, the diagnostic model also matches the untargeted post-1820 acceleration of aggregate real wages (3.11 times the 1710 level by 1881 versus 3.28 in the data), whereas the re-estimated rational model stalls at 1.72 (Allen, 2009). Because diagnostic beliefs reach the allocation only through county investment (Proposition 2), these investment-cycle reversals and transport-network growth profiles cleanly separate θ from construction costs and the persistence of productivity shocks (Table 4; Online Appendix S.E). A belief-neutralization counterfactual (R | DE) then sets $\theta = 0$ while holding all other estimated parameters, the 1710 initial state, statutory project opportunities, and primitive shock paths fixed, and recomputes county investment and the equilibrium path. By 1881, diagnostic expectations expand the aggregate transport stock by +43.79 percent (+54.47 percent in the Midlands–North corridor), raise real output (and real output per worker) by +20.96 percent, and lift the population-weighted worker real wage by +12.97 percent. Larger installed route capacity accelerates structural transformation out of agriculture by −12.9 percentage points (+11.4

percentage points into manufacturing and +1.5 percentage points into services), raises manufacturing frontier productivity by +10.90 percent, and lowers the relative price of manufactures by −13.79 percent and the Stone–Geary price index by −5.97 percent (Table 7 and Figure 9; compare Balboni, 2025; Donaldson, 2018; Donaldson and Hornbeck, 2016; Fajgelbaum and Redding, 2022; Fogel, 1964; Trew, 2020). Decomposing the +20.96 percent output gain across the model's three uninternalized spillover margins shows that direct transit savings account for 25.9 percent of the gain, labor reallocation into wedge-distorted manufacturing accounts for 29.5 percent, and faster spatial knowledge diffusion accounts for 44.7 percent.

This framework bridges quantitative spatial economics and the macroeconomics of endogenous growth and institutional development. Building on Fogel (1964), Krugman (1991), and Donaldson (2018); Donaldson and Hornbeck (2016), a large literature quantifies the market-access gains from transport networks once built (Ahlfeldt et al., 2015; Balboni, 2025; Heblich et al., 2020; Hodgson, 2018; Nagy, 2023; Redding and Turner, 2015), optimal network design (Allen and Arkolakis, 2022; Fajgelbaum and Schaal, 2020), and decentralized transport provision (Bordeu, 2025; Brancaccio et al., 2020; Jiang, 2026) alongside factor-market misallocation (Baqaee and Farhi, 2020; Hornbeck and Rotemberg, 2024; Hsieh and Klenow, 2009; Restuccia and Rogerson, 2008). A parallel literature studies how parliamentary property rights (Bogart, 2014; North, 1990), structural transformation (Broadberry et al., 2014; Crafts and Harley, 1992; Lucas, 2002), coalfield energy and mechanical skill (Allen, 2009; Clark and Jacks, 2007; Kelly et al., 2023), and spatial knowledge diffusion (Berkes et al., 2026; Cinnirella et al., 2024; Lucas, 1988; Hanlon, 2025; Mokyr, 2005; Romer, 1990) shaped Britain's industrial takeoff. By endogenizing county network formation under diagnostic expectations, this paper quantifies how recurrent promoter manias overcame decentralized appropriation wedges to drive British infrastructure accumulation, technology diffusion, and industrialization. Proofs of all propositions are in Appendix A.

I A Simple Example

Consider a two-period regional economy that produces wheat and exports it through a monopoly railway. Exporters' willingness to pay for shipping is governed by the world wheat price Z_t, which follows a stationary AR(1) process $Z_{t+1} = \rho Z_t + \varepsilon_{t+1}$ with innovation ε_{t+1} and persistence $\rho \in (0, 1)$. Inverse shipping demand is linear,

$$P_t(Q_t) = Z_t - bQ_t, \qquad b > 0, \tag{1}$$

where Q_t is the volume shipped. To carry $Q_{t+1} \leq K_{t+1}$ at date $t+1$, the railway must sink irreversible capacity $K_{t+1} \geq 0$ at date t at unit construction cost $c > 0$. Operating costs are zero, and ε_{t+1} has bounded support such that $Z_{t+1} \geq 2bK_{t+1}$ statewise, so installed capacity is fully utilized ($Q_{t+1} = K_{t+1}$; Online Appendix S.A analyzes the general case with ex-post idling and route entry).

A First-Best Capacity

Expected Marshallian social surplus at date t from installing capacity K_{t+1} is $\mathbb{E}_t[S_{t+1}(K_{t+1})] = \mathbb{E}_t\left[\int_0^{K_{t+1}} (Z_{t+1} - bq)\, dq\right] - cK_{t+1} = (\mathbb{E}_t[Z_{t+1}] - c)K_{t+1} - \frac{1}{2}bK_{t+1}^2$. Equating expected marginal social benefit to unit construction cost yields first-best capacity:

$$\mathbb{E}_t\left[P_{t+1}(K_{t+1}^{\mathrm{SP}})\right] = c \quad \Longrightarrow \quad K_{t+1}^{\mathrm{SP}} = \frac{\mathbb{E}_t[Z_{t+1}] - c}{b}. \tag{2}$$

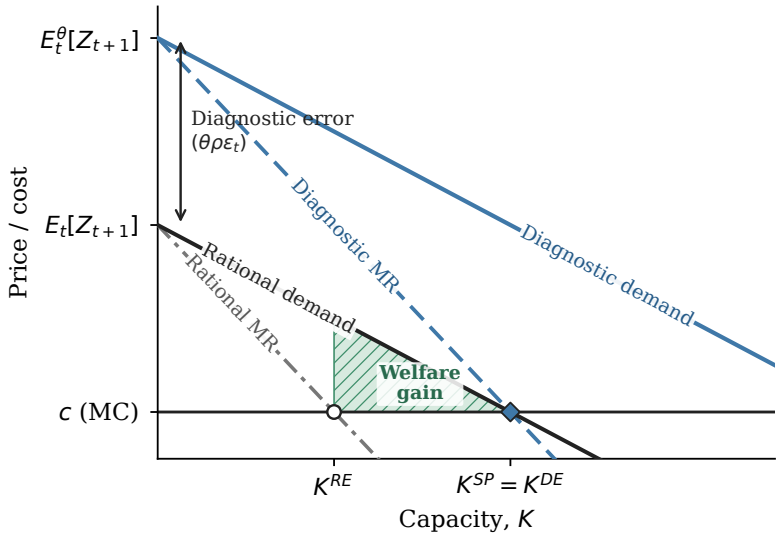


Figure 1: Railways for Wheat Exports

Notes: Linear inverse shipping demand $P = Z - bQ$ and marginal revenue $\mathrm{MR} = Z - 2bQ$. Equating expected marginal revenue rather than marginal social benefit to unit capacity cost c leads rational expectations ($\theta = 0$) to build half the socially efficient capacity ($K^{\mathrm{RE}} = \frac{1}{2}K^{\mathrm{SP}}$), whereas diagnostic overreaction to positive news ($\theta > 0$, $\varepsilon_t > 0$) shifts perceived demand outward and expands sunk capacity toward K^{SP}.

B Monopolist Investment under Rational and Diagnostic Expectations

Following Bordalo et al. (2018), the promoter's diagnostic expectation of Z_{t+1} overweights current news $\mathbb{E}_t[Z_{t+1}] - \mathbb{E}_{t-1}[Z_{t+1}] = \rho\varepsilon_t$:

$$\mathbb{E}_t^\theta[Z_{t+1}] = \mathbb{E}_t[Z_{t+1}] + \theta \underbrace{\left(\mathbb{E}_t[Z_{t+1}] - \mathbb{E}_{t-1}[Z_{t+1}]\right)}_{\text{news}} = \mathbb{E}_t[Z_{t+1}] + \theta\rho\varepsilon_t, \qquad \theta \geq 0, \tag{3}$$

nesting rational expectations at $\theta = 0$. Because the monopolist cannot price discriminate across inframarginal shippers, it equates perceived marginal revenue $\mathbb{E}_t^\theta[\mathrm{MR}_{t+1}] = \mathbb{E}_t^\theta[Z_{t+1}] - 2bK_{t+1}$ to unit cost c:

$$K_{t+1}^{\mathrm{DE}} = \frac{\mathbb{E}_t^\theta[Z_{t+1}] - c}{2b} = \frac{\mathbb{E}_t[Z_{t+1}] + \theta\rho\varepsilon_t - c}{2b}. \tag{4}$$

Under rational expectations ($\theta = 0$), the appropriation wedge between marginal social benefit $Z - bK$ and marginal revenue $Z - 2bK$ halves investment:

$$K_{t+1}^{\mathrm{RE}} = \frac{\mathbb{E}_t[Z_{t+1}] - c}{2b} = \frac{1}{2}K_{t+1}^{\mathrm{SP}}. \tag{5}$$

After positive news $\varepsilon_t > 0$, diagnostic optimism expands sunk capacity by $K_{t+1}^{\mathrm{DE}} - K_{t+1}^{\mathrm{RE}} = \theta\rho\varepsilon_t/(2b) > 0$ and changes expected physical social surplus relative to the rational monopoly benchmark by

$$\mathbb{E}_t\left[S_{t+1}(K_{t+1}^{\mathrm{DE}}) - S_{t+1}(K_{t+1}^{\mathrm{RE}})\right] = \frac{\theta\rho\varepsilon_t}{8b}\left[2\left(\mathbb{E}_t[Z_{t+1}] - c\right) - \theta\rho\varepsilon_t\right]. \tag{6}$$

Equation (6) captures the second-best logic of Lipsey and Lancaster (1956): first-order in $\theta\rho\varepsilon_t > 0$, belief extrapolation acts as a shadow Pigouvian subsidy that narrows the monopoly underinvestment wedge $\mathbb{E}_t[Z_{t+1}] - c$, whereas second-order in $(\theta\rho\varepsilon_t)^2$ it introduces a convex overshooting cost

once capacity exceeds K_{t+1}^{SP}. Every positive shock satisfying $0 < \theta\rho\varepsilon_t < 2(\mathbb{E}_t[Z_{t+1}] - c)$ strictly raises expected social welfare above the rational monopoly outcome—attaining the first best at $\theta\rho\varepsilon_t = \mathbb{E}_t[Z_{t+1}] - c$—and because capacity is irreversible (Arrow, 1968; Dixit and Pindyck, 1994), the network built during the boom persists after beliefs revert.

II Historical Setting and Motivating Evidence

Britain's Industrial Revolution combines three empirical regularities across England and Wales over 1710–1881 that motivate our dynamic spatial framework. First, aggregate growth reallocated labor out of agriculture from a substantial pre-existing industrial base, sorting manufacturing toward the Midlands and Northern coalfields while expanding distribution and transport employment in every county (Broadberry et al., 2014; Crafts and Harley, 1992; Shaw-Taylor and Wrigley, 2014). Second, turnpikes, canals, and railways were promoted by local committees under Parliamentary acts, separating durable route construction from competitive carriage alongside persistent local manufacturing wedges (Bogart, 2014; Hornbeck and Rotemberg, 2024; Trew, 2020; Ward, 1974). Third, major waves of infrastructure building surged after favorable fundamental news and systematically overshot—producing sharp boom–bust reversals in capacity additions and post-boom compression in operating yields (Arnold and McCartney, 2011; Campbell, 2012; Odlyzko, 2010).

A Spatial Structural Transformation and Distribution Labor

To measure regional structural change on a consistent geography, I place county occupational series on fixed 1831 ancient-county boundaries across twelve benchmark dates from 1710 to 1881, combining Keibek (2017) with the CAMPOP national benchmarks (Shaw-Taylor et al., 2019), London registration-county series (Field and Shaw-Taylor, 2019), and the county population weights of Wrigley (2011) (Online Appendix S.B). Occupations follow the Primary–Secondary–Tertiary (PST) classification of Wrigley (2010): primary covers agriculture, forestry, and fishing; secondary groups mining with manufacturing and construction because coalfield geography parallels industrial production (Allen, 2009; Clark and Jacks, 2007); and tertiary comprises transport, communications, dealers, and sellers, isolating the distribution and haulage activities required to move goods across space.

Figures 2 and 3 highlight two features of this transition. Nationally, structural transformation proceeded from a large eighteenth-century industrial base: secondary employment reached half of included employment by 1817, while between 1851 and 1881 the primary share fell from 31 to 19 percent, the secondary share rose from 56 to 62 percent, and the tertiary share expanded from 13 to 18 percent as growing trade volumes required expanding distribution services (Herrendorf et al., 2012; Shaw-Taylor and Wrigley, 2014). Regionally, primary employment remained concentrated in the South and East and secondary industry sorted toward the Midlands and Northern coalfields, whereas distribution and transport employment expanded in every county while remaining thickest around London and the northern industrial hubs (Bogart et al., 2022; Heblich et al., 2020).

B Decentralized Infrastructure Provision, Carriage, and Local Distortions

Unlike the state-planned trunk systems of continental Europe (Milward and Saul, 1973; Smith, 1990), British turnpikes, canals, and railways were promoted decentralizedly by regional committees whose completed links reshaped market access and technology diffusion (Berkes et al., 2026; Bogart, 2014; Mokyr, 2005; Trew, 2020). Across all three eras, statutory organization separated durable route provision from competitive carriage: turnpike trusts, canal companies, and early railway

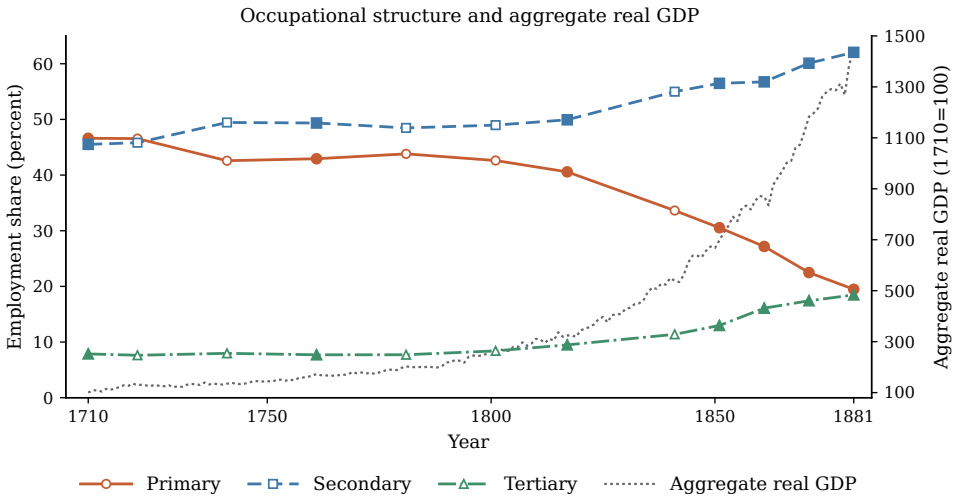


Figure 2: Occupational Structure and Aggregate Real GDP

Notes: Adult male employment shares across primary (excluding mining), secondary (manufacturing, mining, construction), and tertiary (dealers, sellers, transport, communications) sectors: filled markers denote CAMPOP benchmarks (Shaw-Taylor et al., 2019) and open markers denote aligned Keibek (2017) estimates. Right axis: Great Britain real GDP (1710 = 100; Broadberry et al. 2022; Feinstein 1972).

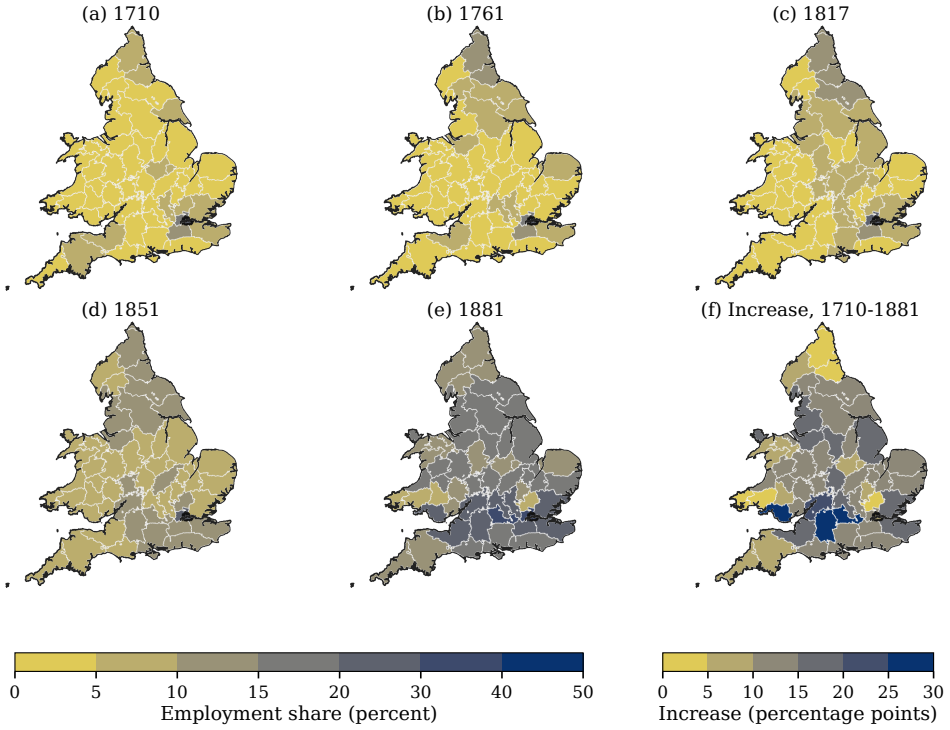


Figure 3: Spatial Expansion of Distribution and Transport Employment, 1710–1881

Notes: Panels (a)–(e) plot county tertiary employment shares at five benchmark dates; Panel (f) plots the 1710–1881 percentage-point increase. County shares through 1851 combine Keibek (2017) and Shaw-Taylor et al. (2019) on 1831 ancient-county boundaries, and 1881 projects the reconstructed latitude-band surface (Online Appendix S.B); the black outline marks London (Field and Shaw-Taylor, 2019).

promoters raised capital to build route capacity and collect tolls within parliamentary-authorized corridors, while independent carriers hired operating labor and vehicles to move freight (Hadfield, 1981; Turnbull, 1987; Ward, 1974).

Network expansion also interacted with persistent local distortions in manufacturing. Desmet, Greif and Parente (2020) document how interregional competition eroded local craft resistance to mechanization unevenly across regions, and Hornbeck and Rotemberg (2024) show that transport improvements generate first-order productivity gains when they reallocate inputs toward wedge-distorted manufacturing producers. Figure 4 plots the cumulative multimode transport stock and annual physical additions over 1710–1881, illustrating the clustered construction waves of the 1790s canal mania and the 1830s–1840s railway booms.

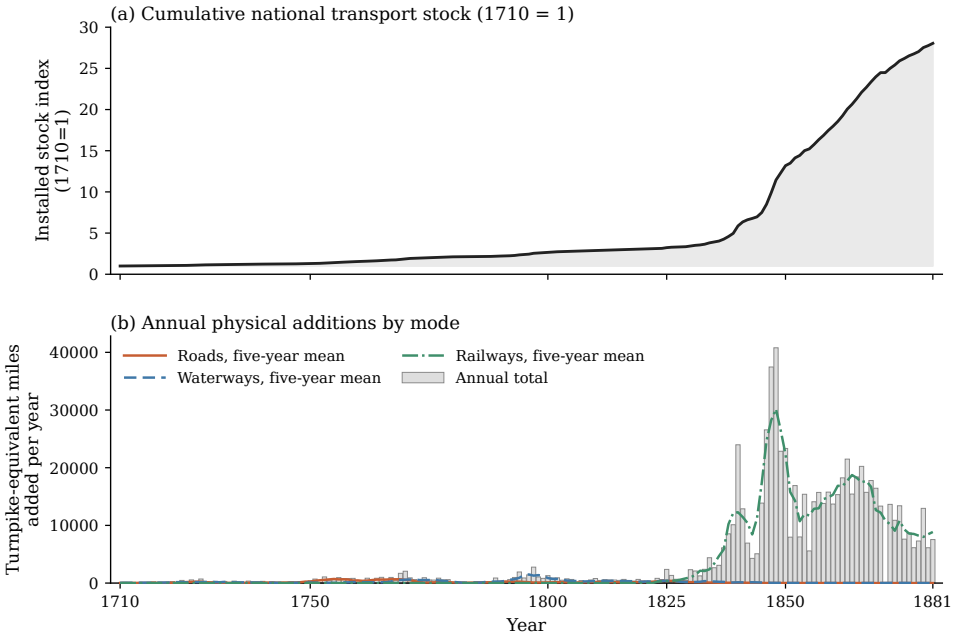


Figure 4: Transport Stock and Installed Additions, 1710–1881

Notes: Turnpikes, waterways, and railways aggregated into turnpike-equivalent miles (road+10×waterway+ 50 × railway). Top panel: cumulative national stock (1710 = 1). Bottom panel: annual physical additions (bars) and five-year moving averages by mode (lines).

C Boom–Bust Reversals and Yield Compression in Infrastructure

Historical accounts of British canal and railway promotion emphasize recurrent cycles of promoter optimism and subsequent disappointment. During the 1830s and the 1845–1848 Railway Mania, prospectuses extrapolated from early trunk successes to advertise 10–15 percent dividends even as realized equity returns averaged under 5 percent and half of Mania-authorized mileage went unbuilt (Odlyzko, 2010). Campbell (2012) shows that weekly railway share prices overreacted to short-run dividend news, while Arnold and McCartney (2011) and Shaen (1847) document systematic cost overruns and post-opening yield shortfalls across early canal and railway companies.

To test whether infrastructure building adjusted smoothly to fundamentals or overshot during booms, let $r^I_{b,t} = (T_{b,t+1} - T_{b,t})/T_{b,t}$ denote the annual growth rate of transport capacity in latitude band b and $c^I_{b,t} = \mathrm{HP}_{6.25}[\log(10^{-3} + r^I_{b,t})]$ its Ravn and Uhlig (2002) cyclical component. Following Jordà (2005) and Bordalo et al. (2018, 2020), I estimate local projections across horizons $h \in$

$\{1, \ldots, 5\}$ for future regional capacity additions $c^I_{b,t+h}$ and aggregate British railway gross operating yields R_{t+h}/K_{t+h} (working receipts over paid-up capital from Mitchell 1988):

$$c^I_{b,t+h} = \alpha_{bh} + \delta_{th} + \rho_h c^I_{b,t} + u_{b,t+h}, \qquad \frac{R_{t+h}}{K_{t+h}} = \alpha^R_h + \beta^R_h \Delta \ln K_t + u^R_{t+h}. \tag{7}$$

Under rational construction smoothing, a building surge at date t predicts positive future additions above trend ($\rho_h > 0$). Under boom-time overshooting, excess capacity depresses subsequent additions below trend ($\rho_h < 0$) and compresses subsequent operating yields ($\beta^R_h < 0$). Estimating ρ_h directly on the future level $c^I_{b,t+h}$ places no $-c^I_{b,t}$ term on the left-hand side, avoiding mechanical differencing (Jordà, 2005; López-Salido et al., 2017).

Table 1: Local Projections of Infrastructure Investment and Railway Yields

	(1) $h=1$	(2) $h=2$	(3) $h=3$	(4) $h=4$	(5) $h=5$
Panel A. Dependent variable: Regional investment cycle $c^I_{b,t+h}$ (25 latitude bands, 1710–1880)					
Investment cycle $c^I_{b,t}$ (TWFE)	−0.221	−0.215	−0.121	−0.028	−0.011
	(0.019)	(0.020)	(0.020)	(0.020)	(0.019)
Investment cycle $c^I_{b,t}$ (2SLS)	−0.273	−0.202	−0.084	−0.088	−0.004
	(0.054)	(0.057)	(0.054)	(0.056)	(0.028)
Band and year fixed effects	Yes	Yes	Yes	Yes	Yes
First-stage F-statistic (2SLS)	175.6	174.8	174.4	173.2	172.3
Observations	4,250	4,225	4,200	4,175	4,150
Panel B. Dependent variable: Railway gross operating yield R_{t+h}/K_{t+h} (percent, 1843–1881)					
Log paid-up capital growth $\Delta \ln K_t$ (OLS)	−13.00	−13.71	−13.65	−12.49	−11.15
	(2.23)	(2.06)	(2.31)	(2.37)	(2.19)
R^2	0.46	0.52	0.51	0.45	0.40
Observations	38	37	36	35	34

Notes: Local projections (7) at horizons $h = 1, \ldots, 5$ years (Jordà, 2005). Panel A estimates regional regressions across 25 latitude bands (1710–1880) by TWFE and by 2SLS using shift-share instruments $Z_{b,t} = (e^{\text{coal}}_b \bar{c}^{I,\text{coal}}_{-b,t},\ \pi_{M,b,1710}\bar{c}^{I,M}_{-b,t},\ (0.5 + e^{\text{coal}}_b)\bar{c}^I_{-b,t})'$ that interact predetermined GIS coalfield exposure e^{coal}_b (Bogart et al., 2022) and 1710 manufacturing shares $\pi_{M,b,1710}$ with leave-one-out regional addition cycles; band-clustered standard errors are in parentheses. Panel B regresses British railway gross working receipts over paid-up capital R_{t+h}/K_{t+h} (percent) on annual log paid-up capital growth $\Delta \ln K_t$ over 1843–1881 (Mitchell, 1988), with Newey–West HAC standard errors in parentheses.

Table 1 documents sharp post-boom reversals in both physical construction and realized operating yields. In Panel A, a one-unit cyclical surge in regional capacity additions is followed by below-trend additions over the next three years ($\rho_1 = -0.221$, SE $= 0.019$; $\rho_2 = -0.215$, SE $= 0.020$; $\rho_3 = -0.121$, SE $= 0.020$). Instrumenting regional building with predetermined coalfield and 1710 manufacturing exposure interacted with leave-one-out regional addition cycles (first-stage $F > 172$) yields closely matching reversals ($\rho^{\text{IV}}_1 = -0.273$, SE $= 0.054$; $\rho^{\text{IV}}_2 = -0.202$, SE $= 0.057$), confirming that fundamental-driven booms overshoot and contract.[3] In Panel B, railway paid-up capital surges $\Delta \ln K_t$ predict lower subsequent gross operating yields across all five horizons ($\beta^R_1 = -13.00$ to $\beta^R_5 = -11.15$, $R^2 = 0.40$–0.52), consistent with the 34 percent shortfall between prospectus and realized net traffic yields documented in Shaen (1847)'s parliamentary audit.

[3]Online Appendix S.B (Table S.B.1) shows that these reversals are robust to two-way clustering by latitude band and year, leave-corridor-out ($|k - b| \geq 3$) shift-share instruments, one-sided backward-looking detrending, untransformed level and inverse-hyperbolic-sine addition rates, unfiltered cumulative changes, and separate pre-railway (1710–1829) and railway (1830–1880) subsamples.

Taken together, medium-horizon construction reversals ($\rho_h < 0$), predictable post-boom yield compression ($\beta_h^R < 0$), and systematic shortfalls relative to prospectus forecasts discriminate against standard rational mechanisms. First, under multi-period time-to-build and convex adjustment costs (Hayashi, 1982; Kalouptsidi, 2014; Kydland and Prescott, 1982), multi-year gestation lags generate positive short-run completion momentum ($\rho_1 = +0.067$ during the railway era and $\rho_1^{1s} = +0.159$ under one-sided detrending; Online Appendix Table S.B.1), but additions return to trend once projects finish rather than dropping below trend at medium horizons ($\rho_2, \rho_3 < 0$). Nor can known construction lags explain why date-t capital growth $\Delta \ln K_t \in \mathcal{F}_t$ predicts five years of operating-yield compression in Panel B or why realized yields systematically missed promoters' own parliamentary prospectuses (Shaen, 1847). Second, under Bayesian learning and irreversible real options (Arrow, 1968; Bernanke, 1983; Dixit and Pindyck, 1994; Jovanovic, 1982; Pástor and Veronesi, 2009b), uncertainty raises the option value of waiting—delaying and smoothing entry rather than bunching it into booms—while martingale belief updating implies gradual continuation ($\rho_h > 0$) and unpredictable ex post return errors conditional on date-t investment. Third, in the aggregate time series, cyclical additions load strongly on current and lagged sectoral output cycles ($b_0 = 2.95$, $b_1 = 1.58$) but have zero predictive power for next-year output ($b_{\text{lead}} = -0.005$, SE $= 0.004$), indicating that promoters extrapolated recent industrial expansions rather than acting on private foresight of future fundamentals (Cochrane, 1991). Finally, year fixed effects δ_{th} in Panel A absorb aggregate interest-rate and risk-premium fluctuations so that the 2SLS reversals reflect cross-regional overreaction to real industrial news within each year, while Mania-era prospectuses advertised 10–15 percent dividends—above, not below, historical norms (Campbell, 2012; Odlyzko, 2010).

III The Model

A The Spatial Economy

Time is discrete, $t \in \{0, 1, 2, \dots\}$. Building on the continuous-space geography of Lucas and Rossi-Hansberg (2002) and Rossi-Hansberg (2005) and its extension to spatial growth and endogenous transport by Desmet and Rossi-Hansberg (2014) and Trew (2020), the economy occupies the unit interval $\ell \in [0, 1]$, partitioned into J county jurisdictions $\mathcal{C}_j = [c_{j-1}, c_j]$, with output, employment, and route infrastructure measured as densities per unit of land. Three sectors operate across space: agriculture (A) and manufacturing (M), which produce the two consumption goods, and transport and distribution services (T). Competitive land leasing assigns each site $\ell \in [0, 1]$ to a single goods sector ($e_A(\ell, t) + e_M(\ell, t) = 1$, $e_i(\ell, t) \in \{0, 1\}$), while competitive transport carriers operate alongside goods producers within each county.

A continuum of worker households of mass $\bar{L}_t$ supplies one unit of labor inelastically and holds equal shares in the national portfolio of land leases, producer net profits, carrier capacity rents, and factor-wedge rebates. Under constant participation, England and Wales population P_t^{EW} determines the household mass (normalized to 100 in 1710):

$$\bar{L}_t = 100 \frac{P_t^{EW}}{P_{1710}^{EW}}, \tag{8}$$

with each household receiving $1/\bar{L}_t$ of aggregate nonlabor income. In each county $j \in \{1, \dots, J\}$, a parliamentary-authorized infrastructure company chooses completion intensity x_{jt} to maximize expected discounted land rents inside $\mathcal{C}_j$ net of manufactured construction expenditure.

As in dynamic spatial general equilibrium frameworks (Kleinman et al., 2023), each period t unfolds in two stages conditional on inherited knowledge $A_{it}(\ell)$, route capacity $T_t(\ell)$, and aggregate

productivity gaps $\mathbf{z}_t$, separating forward-looking state transitions from within-period spot market clearing:

1. *Pre-discovery commitments:* County companies commit $\mathbf{x}_t \in \mathcal{X}_t$; goods entrants announce pure sector $e_i(\ell, t)$, research probability $\phi_i(\ell, t)$, innovation bill Ψ_i, and hired production labor $L_i(\ell, t)$ at a fixed local contract wage $w(\ell, t)$; carriers commit operating labor $L_T(\ell, t)$ at $w(\ell, t)$; and households choose residence density $n(\ell, t)$.

2. *Post-discovery spot clearing and transitions:* Local innovation gains $G_{it}(\ell, \omega)$ realize; output utilization $Y_i(\ell, t, \omega)$, county tolls $\tau_j(t, \omega)$, spot prices $p_i(\ell, t, \omega)$, trade flows $H_i(\ell, t, \omega)$, and consumption $c_i(\ell, t, \omega)$ clear markets statewise; and discoveries and completed projects update $A_{i,t+1}(\ell)$ and $T_{t+1}(\ell)$.

Table 2 summarizes the environment; Subsections III.B–III.G derive closed-form rules that reduce the equilibrium to a coupled spatial operator and a dynamic county marginal-q game.

Table 2: Economic Environment

	Households
Lifetime utility	$W_t = \max_{\{\ell_s, c_A(s), c_M(s)\}_{s \geq t}} \mathbb{E}_t^\theta[\sum_{s=t}^{\infty} \beta^{s-t} U(c_A(s), c_M(s))]$
Period utility	$U(c_A, c_M) = (c_A - \gamma)^\eta c_M^{1-\eta}$
Budget	$p_A c_A + p_M c_M = w + b_t(\omega)$
Nonlabor income	$b_t = \bar{L}_t^{-1} \int_0^1 (R - p_M I + \Pi_F + \Pi_T + \Pi_\psi)\, d\ell$
	Production and Innovation
Bid rent	$R_i = \max_{\substack{0 \leq \phi_i < 1 \\ \bar{L}_i \geq 0}} \{\mathbb{E}_t^\theta[(p_i - r_i\tau_j)_+ Y_i] - (1+\psi_i) w L_i - \Psi_i(\phi_i; w, \ell)\}$
Production technology	$Y_i = e_i Z_i G_i L_i^{\alpha_i}$
Innovation expenditure	$\Psi_i(\phi_i; w, \ell) = w\left[f_i(\ell) + \frac{\chi_i}{1-\phi_i}\right]$ for $\phi_i > 0$, $\quad \Psi_i(0; w, \ell) = 0$
Research outcome	$\Pr(G_i > g \mid \phi_i) = \phi_i g^{-a_i}$, $\quad g \geq 1$
Knowledge after innovation	$A_{it}^+ = A_{it}[1 + e_{it}(G_{it} - 1)]$
	Transport Services
Operating profits	$\max_{L_T \geq 0} \mathbb{E}_t^\theta[\tau_j Y_T - w L_T]$
Transport technology	$Y_T = Z_T L_T^{\alpha_T} T^{1-\alpha_T}$
County capacity and demand	$K_j = \int_{\mathcal{C}_j} Y_T\, d\ell$, $\quad D_j = \int_{\mathcal{C}_j} \sum_i r_i (Y_i + \lvert H_i \rvert)\, d\ell$
	County Infrastructure Companies
Value function	$V_{jt}^\theta = \max_{0 \leq x_j \leq \bar{x}_{jt}} \{R_{jt} + \beta \mathbb{E}_t^\theta[V_{j,t+1}^\theta]\}$
Net rents	$R_{jt} = \mathbb{E}_t^\theta \int_{\mathcal{C}_j} [R_t - p_{Mt} I_t(\ell; \mathbf{x}_t)]\, d\ell$
Installed capacity	$T_{t+1} = T_t + \sum_j x_{jt} \iota_{jt}$
	State Transitions
Productivity	$Z_{it} = A_{it} e^{z_{it}} / \mathbb{E}_\pi[e^{z_i}]$, $\quad i \in \{A, M, T\}$
Sectoral shocks	$\mathbf{z}_{t+1} \mid \mathbf{z}_t \sim \mathrm{TN}_{\mathcal{B}}(\boldsymbol{\rho} \mathbf{z}_t, \boldsymbol{\Sigma})$
Available frontier	$F_{it}(\ell) = \sup_{r \in [0,1]} e^{-\delta \lvert \ell - r \rvert} A_{it}^+(r)$
Adoption	$A_{i,t+1} = A_{it}^+ + \left[1 - e^{-\lambda_0 (T_t / T_{t_0})^{1-\alpha_T}}\right](F_{it} - A_{it}^+)$
	Market Clearing
Residence	$\mathbb{E}_t^\theta[(w + b - \gamma p_A)/\mathcal{P}] = \bar{u}$
Goods	$\partial_\ell H_i = Y_i - n c_i - e_i \Psi_i / p_i - I_i - \kappa_0 e^{-T_t} \lvert H_i \rvert$
Labor and population	$n = \sum_i e_i L_i + L_T$, $\quad \int_0^1 n\, d\ell = \bar{L}_t$
Services	$D_j \leq K_j$, $\quad \tau_j \geq 0$, $\quad \tau_j (K_j - D_j) = 0$

Notes: Local and date arguments are suppressed; $i \in \{A, M\}$ unless stated otherwise, and j indexes the county. $e_i \in \{0, 1\}$ indicates the highest-bidding goods sector with land rent $R = \max_i R_i$; $f_A(\ell) = f_A$ and $f_M(\ell) = f_M + \xi(\ell)$; construction uses manufactured goods ($I_A = 0$, $I_M = I$). Borders are closed and the numeraire is $p_A(1, t) = 1$. Remaining definitions are in the text.

B State and Beliefs

At the start of date t, the persistent spatial state consists of bounded, positive, measurable knowledge profiles $A_{At}(\ell)$ and $A_{Mt}(\ell)$ and installed route capacity $T_t(\ell)$, observed alongside aggregate sectoral productivity gaps $\mathbf{z}_t = (z_{At}, z_{Mt}, z_{Tt})'$ and the predetermined demographic, coalfield energy, manufacturing-wedge, and statutory project schedules in $\mathcal{E}_t \subset \mathcal{F}_t$. Only the active sector at ℓ conducts research there, whereas both sectors adopt discoveries from the spatial frontier; infrastructure built at date t enters operating capacity at $t+1$ and accelerates knowledge adoption from $t+1$ onward.

Aggregate productivity gaps follow a stationary first-order Markov process on a compact sym-

metric box $\mathcal{B}$:

$$\mathbf{z}_{t+1} \mid \mathbf{z}_t \sim \mathrm{TN}_{\mathcal{B}}(\boldsymbol{\rho}\mathbf{z}_t, \boldsymbol{\Sigma}), \tag{9}$$

where $\mathrm{TN}_{\mathcal{B}}$ is a multivariate Gaussian law truncated to $\mathcal{B}$ ($\pm c$ stationary untruncated standard deviations around zero, with $c = 3$ in the benchmark and $c = 4$ for robustness; Online Appendix S.C.1), $\boldsymbol{\rho} = \mathrm{diag}(\rho_A, \rho_M, \rho_T)$, and $\boldsymbol{\Sigma}$ is the innovation covariance matrix. The kernel has a unique stationary distribution π on $\mathcal{B}$, and effective productivity (18) is normalized by $\mathbb{E}_\pi[e^{z_i}]$ so that $e^{z_{it}}/\mathbb{E}_\pi[e^{z_i}]$ has unit stationary physical mean. Write $f(\mathbf{z}' \mid \mathbf{z})$ for this physical kernel and $M(\mathbf{m})$ for the conditional mean on $\mathcal{B}$ of a Gaussian centered at $\mathbf{m}$.

Belief formation follows the diagnostic expectations operator of Bordalo et al. (2018, 2020), which formalizes Kahneman and Tversky (1972)'s representativeness heuristic and underpins macroeconomic models of sentiment-driven cycles (Bianchi et al., 2024; Bordalo et al., 2026; L'Huillier et al., 2023; Maxted, 2023). Relative to the no-news reference forecast $\overline{\mathbf{z}}_t = \mathbb{E}_{t-1}[\mathbf{z}_t] = M(\boldsymbol{\rho}\mathbf{z}_{t-1})$, observed news $\boldsymbol{\varepsilon}_t = \mathbf{z}_t - \overline{\mathbf{z}}_t$ tilts perceived transition probabilities toward states whose likelihood increased most in light of $\boldsymbol{\varepsilon}_t$, weighted by the representativeness ratio $\Lambda_t(\mathbf{z}') = f(\mathbf{z}' \mid \mathbf{z}_t)/f(\mathbf{z}' \mid \overline{\mathbf{z}}_t)$:

$$\mathbb{E}_t^\theta[X] = \frac{\mathbb{E}_t[X\Lambda_t(\mathbf{z}_{t+1})^\theta]}{\mathbb{E}_t[\Lambda_t(\mathbf{z}_{t+1})^\theta]}, \qquad \theta \geq 0, \tag{10}$$

where $\theta = 0$ nests rational expectations and $\theta > 0$ indexes belief extrapolation. Because $\mathbf{z}_t$ is observed at the start of date t and local Bernoulli–Pareto discoveries are independent of $\mathbf{z}_{t+1}$, the factor $\Lambda_t(\mathbf{z}_{t+1})^\theta$ cancels from (10) for any within-period payoff conditional on $(\Omega_t, \mathbf{x}_t)$. Exponentiating the Gaussian kernel yields the closed-form diagnostic one-step law and recursive forecast reference (Online Appendix S.C.1):

$$\mathbf{z}_{t+1} \overset{\theta}{\sim} \mathrm{TN}_{\mathcal{B}}(\mathbf{m}_t^\theta, \boldsymbol{\Sigma}), \qquad \mathbf{m}_t^\theta = \boldsymbol{\rho}\mathbf{z}_t + \theta\boldsymbol{\rho}(\mathbf{z}_t - \overline{\mathbf{z}}_t), \tag{11}$$

$$\overline{\mathbf{z}}_{t+1}^{\text{forecast}} = M(\mathbf{m}_t^\theta), \qquad \boldsymbol{\varepsilon}_{t+1}^{\text{forecast}} = \mathbf{z}_{t+1} - M(\mathbf{m}_t^\theta), \tag{12}$$

whereas in physical time at date $t+1$, the observed reference resets to $\overline{\mathbf{z}}_{t+1} = M(\boldsymbol{\rho}\mathbf{z}_t)$. Equation (11) delivers the two empirical regularities documented in Section II: forecasts overreact to current news with location shift $(1+\theta)\boldsymbol{\rho}\boldsymbol{\varepsilon}_t$, and subsequent realizations systematically disappoint after positive news ($\mathbf{m}_t^\theta - \boldsymbol{\rho}\mathbf{z}_t = \theta\boldsymbol{\rho}\boldsymbol{\varepsilon}_t$). The aggregate state is $\Omega_t = (A_{At}, A_{Mt}, T_t, \mathbf{z}_t, \overline{\mathbf{z}}_t, t)$ (Figure 5).

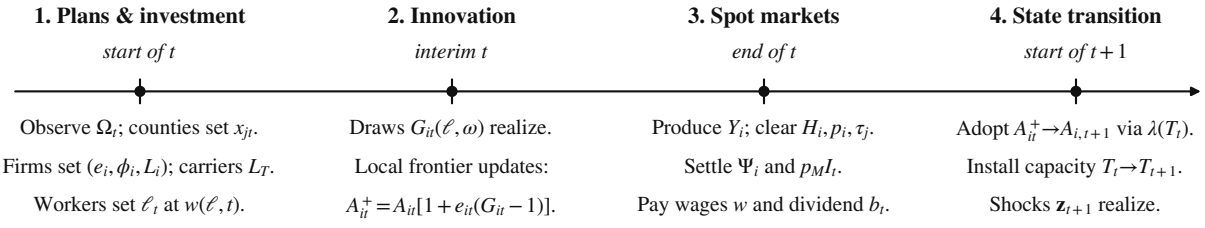


Figure 5: Within-Period Timing

Notes: County investment x_{jt}, producer plans (e_i, ϕ_i, L_i), carrier employment L_T, and worker residence ℓ_t at local contract wage $w(\ell, t)$ are committed before idiosyncratic discoveries $G_{it}(\ell, \omega)$ realize. Spot goods and service markets clear ex post, and completed capacity T_{t+1} and diffused knowledge $A_{i,t+1}$ enter at date $t+1$.

C Workers

Workers live where they work, supply one unit of labor inelastically, and hold equal shares in the national portfolio of land leases, goods producers, transport firms, and factor-wedge rebates,

receiving common nonlabor income b_t in addition to the local contract wage $w(\ell, t)$. Goods are nonstorable, and households neither save nor borrow. Each period t, a worker commits residence $\ell_t \in [0, 1]$ before private discoveries at the fixed contract wage $w(\ell, t)$ and chooses spot consumption $(c_A(\ell, t), c_M(\ell, t))$ ex post to solve

$$W_t = \max_{\{\ell_s, c_A(s), c_M(s)\}_{s \geq t}} \mathbb{E}_t^\theta \left[\sum_{s=t}^{\infty} \beta^{s-t} U\big(c_A(s), c_M(s)\big) \right] \\ \text{s.t. } p_A(\ell_s, s) c_A(s) + p_M(\ell_s, s) c_M(s) = w(\ell_s, s) + b_s, \quad c_A(s) \geq \gamma,\ c_M(s) \geq 0, \tag{13}$$

with nonhomothetic Stone–Geary period utility (Fajgelbaum and Redding, 2022; Herrendorf et al., 2014; Kongsamut et al., 2001):

$$U\big(c_A(\ell, t), c_M(\ell, t)\big) = (c_A(\ell, t) - \gamma)^\eta c_M(\ell, t)^{1-\eta}, \qquad \eta \in (0, 1), \quad \gamma > 0. \tag{14}$$

For $w + b_t \geq \gamma p_A$, spot optimization yields the Marshallian demands, marginal price index, and realized indirect utility:

$$c_A = \gamma + \frac{\eta(w + b_t - \gamma p_A)}{p_A}, \qquad c_M = \frac{(1-\eta)(w + b_t - \gamma p_A)}{p_M}, \\ \mathcal{P}(p_A, p_M) = \frac{p_A^\eta p_M^{1-\eta}}{\eta^\eta (1-\eta)^{1-\eta}}, \qquad u_t(\ell) = \frac{w(\ell, t) + b_t - \gamma p_A(\ell, t)}{\mathcal{P}(p_A(\ell, t), p_M(\ell, t))}. \tag{15}$$

The subsistence floor $\gamma > 0$ generates Engel's law, shifting expenditure toward manufacturing as real income rises. Because residence is committed before private discoveries, workers evaluate locations by expected current indirect utility $\mathbb{E}_t^\theta[u_t(\ell)]$.

PROPOSITION 1: *If lifetime utility is finite, then, given the current price, wage and portfolio-income schedules, a worker's optimal dynamic residence choice maximizes expected current-period indirect utility.*[4] [Proof →]

Because annual relocation is costless and portfolio claims do not depend on residence, continuation utility $\mathbb{E}_t^\theta[W_{t+1}]$ is invariant to current location, equalizing expected indirect utility across occupied sites:

$$\mathbb{E}_t^\theta\big[u_t(\ell)\big] = \bar{u}_t \qquad \text{where } n(\ell, t) > 0 \quad \big(\mathbb{E}_t^\theta[u_t(\ell)] \leq \bar{u}_t \text{ elsewhere}\big), \tag{16}$$

$$w(\ell, t) = \frac{\bar{u}_t - \mathbb{E}_t^\theta\big[(b_t - \gamma p_A(\ell, t))/\mathcal{P}(p_A(\ell, t), p_M(\ell, t))\big]}{\mathbb{E}_t^\theta\big[1/\mathcal{P}(p_A(\ell, t), p_M(\ell, t))\big]}. \tag{17}$$

D Production, Research, and Land

Competitive entrants in sector $i \in \{A, M\}$ bid for one-period pure site leases at $\ell \in [0, 1]$.[5] Effective productivity combines permanent local knowledge $A_i(\ell, t)$ (or constant carrier frontier $A_T > 0$) with the aggregate productivity factor, and realized output Y_i uses committed labor $L_i(\ell, t)$ under diminishing returns:

$$Z_i(\ell, t) = A_i(\ell, t) e^{z_{it}} / \mathbb{E}_\pi[e^{z_i}], \quad i \in \{A, M\}, \qquad Z_T(t) = A_T e^{z_{Tt}} / \mathbb{E}_\pi[e^{z_T}], \tag{18}$$

$$Y_i(\ell, t, \omega) = e_i(\ell, t) Z_i(\ell, t) G_i(\ell, t, \omega) L_i(\ell, t)^{\alpha_i}, \qquad i \in \{A, M\}, \quad 0 < \alpha_A < \alpha_M < 1, \tag{19}$$

[4] Proofs of the propositions are in Appendix A.

[5] Transport routes occupy narrow corridors and do not crowd out productive land.

where $\alpha_A < \alpha_M$ reflects the higher land intensity of agriculture relative to manufacturing (Valentinyi and Herrendorf, 2008). Entrants announce (e_i, ϕ_i, L_i) taking state-contingent spot prices $p_i(\ell, t, \omega)$, county tolls $\tau_{j(\ell)}(t, \omega)$, and the local contract wage $w(\ell, t)$ as given, earning the net unit price $p_i(\ell, t, \omega) - r_i\tau_{j(\ell)}(t, \omega)$ ex post.

Manufacturing firms face a predetermined local operating-labor wedge,

$$\psi_A(\ell) = 0, \qquad \psi_M(\ell) = \bar{\psi}_M \omega_M(\ell), \qquad \bar{\psi}_M \geq 0, \quad \omega_M(\ell) > 0, \tag{20}$$

paying $(1 + \psi_M(\ell))w(\ell, t)$ per hired production worker while agriculture is undistorted ($\psi_A = 0$) and wedge receipts are rebated lump-sum to households. Holding $\omega_M(\ell)$ fixed makes transport improvements a source of first-order allocative efficiency gains as labor reallocates toward distorted industrial producers (Baqaee and Farhi, 2020; Hornbeck and Rotemberg, 2024).

Before private discoveries, an entrant in sector $i \in \{A, M\}$ chooses a research success probability $\phi_i \in [0, 1)$. Building on the Pareto idea-flow representation of Kortum (1997) and Eaton and Kortum (2002), a successful draw yields a Pareto productivity multiplier with tail exponent $a_i > 1/(1 - \alpha_i)$:

$$\Pr(G_i = 1 \mid \phi_i) = 1 - \phi_i, \tag{21}$$

$$\Pr(G_i > g \mid \phi_i) = \phi_i g^{-a_i}, \qquad g \geq 1, \quad a_i > 1. \tag{22}$$

Discoveries are spatially correlated. A success field $X^s_{it}(\ell)$ and a quality field $X^g_{it}(\ell)$ are stationary standard Gaussian Ornstein–Uhlenbeck processes on $[0, 1]$, independent of each other, across sectors and dates, and of aggregate productivity gaps, with correlation length $d_0 > 0$; with Φ the standard normal distribution function, an entrant choosing ϕ at ℓ draws

$$\begin{aligned} G_i(\phi; \ell, \omega) &= 1 + \mathbf{1}_{\{\Phi(X^s_{it}(\ell)) < \phi\}} \left[\Phi(X^g_{it}(\ell))^{-1/a_i} - 1\right], \\ \mathrm{Cov}(X^h_{it}(\ell), X^h_{it}(r)) &= e^{-|\ell - r|/d_0}, \qquad h \in \{s, g\}. \end{aligned} \tag{23}$$

Because $\Phi(X^h_{it}(\ell))$ is uniform at every site, (23) reproduces (21)–(22) at each ℓ, while neighboring sites draw correlated discoveries. All research intensities share one probability space: $G_i(\phi; \ell, \omega)$ is nondecreasing in ϕ, and $G_i(\phi_n; \ell, \omega) \to G_i(\phi; \ell, \omega)$ almost surely whenever $\phi_n \to \phi$.

Innovation requires a wage-indexed monetary expenditure committed before discovery and settled in the spot market by purchasing own-sector output locally:

$$\Psi_i(\phi_i; w, \ell) = \begin{cases} w\left[f_i(\ell) + \dfrac{\chi_i}{1 - \phi_i}\right] & \text{if } \phi_i \in (0, 1), \\ 0 & \text{if } \phi_i = 0, \end{cases} \tag{24}$$

$$q^R_i(\omega) = \frac{\Psi_i(\phi_i; w, \ell)}{p_i(\omega)}, \tag{25}$$

where $f_A(\ell) = f_A \geq 0$ and $f_M(\ell) = f_M + \xi(\ell) \geq 0$ combines fixed participation cost $f_i \geq 0$ with predetermined manufacturing energy overhead $\xi(\ell) = \omega_E e(\ell) \geq 0$ ($\omega_E \geq 0$ scaling the standardized coalfield energy exposure schedule $e(\ell)$ of Allen 2009), and $\chi_i > 0$ governs research convexity. Gross marketed output Y_i includes research goods, using local distribution services $r_i Y_i$ once, while only the knowledge transition carries discoveries to future periods.

Firm optimization and innovation threshold.— Risk-neutral firms measure payoffs in the numeraire $p_A(1, t) = 1$ and hold neither future land claims nor patents. Taking state-contingent goods prices $p_i(\omega)$, county tolls $\tau_{j(\ell)}(\omega)$, and the local contract wage w as given, a producer with hired labor L_i and discovery $G_i(\omega)$ chooses spot output utilization to maximize operating surplus:

$$\max_{0 \leq Y \leq Z_i G_i L_i^{\alpha_i}} \left\{(p_i - r_i\tau_{j(\ell)})Y - (1 + \psi_i)wL_i\right\} = (p_i - r_i\tau_{j(\ell)})_+ Z_i G_i L_i^{\alpha_i} - (1 + \psi_i)wL_i, \tag{26}$$

where $[x]_+ = \max\{x, 0\}$, so full capacity $Y_i = e_i Z_i G_i L_i^{\alpha_i}$ is operated whenever $p_i(\omega) > r_i \tau_{j(\ell)}(\omega)$. Writing expected net unit revenue per efficiency unit as

$$m_i(\phi_i; \ell) = \mathbb{E}_t^\theta \Big[\big(p_i(\ell, \omega) - r_i \tau_{j(\ell)}(\omega)\big)_+ G_i(\phi_i; \ell, \omega) \Big], \tag{27}$$

the entrant maximizes expected land-bid surplus:

$$R_i(\ell) = \max_{\substack{0 \le \phi_i < 1 \\ L_i \ge 0}} \Big\{ Z_i(\ell) m_i(\phi_i; \ell) L_i^{\alpha_i} - (1 + \psi_i(\ell)) w(\ell) L_i - \Psi_i(\phi_i; w(\ell), \ell) \Big\}. \tag{28}$$

First-order optimization in L_i yields the closed-form hiring rule and profiled scalar innovation problem:

$$L_i^*(\phi_i; w, \ell) = \left[\frac{\alpha_i Z_i(\ell) m_i(\phi_i; \ell)}{(1 + \psi_i(\ell)) w(\ell)} \right]^{\frac{1}{1-\alpha_i}}, \tag{29}$$

$$R_i(w, \ell) = \max_{\phi_i \in [0,1)} \Big\{ \Lambda_i(\ell) w(\ell)^{-\frac{\alpha_i}{1-\alpha_i}} m_i(\phi_i; \ell)^{\frac{1}{1-\alpha_i}} - \Psi_i(\phi_i; w(\ell), \ell) \Big\}, \tag{30}$$

with $L_i^*(\phi_i; w, \ell) = 0$ if $m_i(\phi_i; \ell) = 0$ and $\Lambda_i(\ell) = (1 - \alpha_i) Z_i(\ell) [\alpha_i Z_i(\ell)/(1 + \psi_i(\ell))]^{\alpha_i/(1-\alpha_i)}$. On $\phi_i \in (0, 1)$, Ψ_i is strictly convex with $\partial^2_{\phi\phi} \Psi_i = 2w(\ell)\chi_i/(1 - \phi_i)^3 > 0$ and diverges as $\phi_i \uparrow 1$. By the envelope theorem, an interior research optimum $\tilde{\phi}_i(w, \ell) \in (0, 1)$ satisfies

$$Z_i(\ell) L_i^*(\tilde{\phi}_i; w, \ell)^{\alpha_i} \partial_\phi m_i(\tilde{\phi}_i; \ell) = \frac{w(\ell)\chi_i}{\big(1 - \tilde{\phi}_i(w, \ell)\big)^2}, \tag{31}$$

and for $\chi_i > \bar{\chi}_{it}$ the active objective is strictly concave on $(0, 1)$ with at most one interior maximizer. Dividing (30) by $w(\ell)$ with baseline profitability index $\pi_i^0(w, \ell) \equiv \Lambda_i(\ell) [m_i(0; \ell)/w(\ell)]^{1/(1-\alpha_i)}$ and discovery gain $\Gamma_i(\phi_i; \ell) \equiv [m_i(\phi_i; \ell)/m_i(0; \ell)]^{1/(1-\alpha_i)} - 1$ gives

$$\frac{R_i(w, \ell)}{w(\ell)} = \pi_i^0(w, \ell) + \max\left\{ 0, \max_{\phi_i \in (0,1)} \left[\pi_i^0(w, \ell) \Gamma_i(\phi_i; \ell) - f_i(\ell) - \frac{\chi_i}{1 - \phi_i} \right] \right\}. \tag{32}$$

Comparing $\tilde{\phi}_i(w, \ell)$ with zero research defines a unique reservation threshold $\underline{\pi}_i(\ell) > 0$: innovation is active ($\phi_i^*(w, \ell) = \tilde{\phi}_i(w, \ell) > 0$) if and only if $\pi_i^0(w, \ell) > \underline{\pi}_i(\ell)$, and zero otherwise. Transport improvements $T_t(\ell)$ raise net unit prices $p_i(\ell, \omega) - r_i \tau_{j(\ell)}(\omega)$ while coalfield proximity lowers $\xi(\ell)$, pushing $\pi_M^0(w, \ell)$ above $\underline{\pi}_M(\ell)$ in industrializing counties. Since $\pi_i^0(w, \ell) \propto w(\ell)^{-1/(1-\alpha_i)}$, higher contract wages weakly lower $\phi_i^*(w, \ell)$ and strictly lower $L_i^*(\phi_i^*(w, \ell); w, \ell)$ (Appendix A.6.1).

Land competition and spatial sorting.— Competitive bidding assigns each site to the sector with the higher expected bid rent:

$$R(\ell) = \max\{R_A(\ell),\, R_M(\ell)\}, \qquad e_i(\ell) = \begin{cases} 1 & \text{if } R_i(\ell) = R(\ell), \\ 0 & \text{otherwise.} \end{cases} \tag{33}$$

Because $\alpha_A < \alpha_M$ and $\Lambda_i(\ell) \propto Z_i(\ell)^{1/(1-\alpha_i)} (1 + \psi_i(\ell))^{-\alpha_i/(1-\alpha_i)}$, an increasing elasticity- and wedge-adjusted comparative-advantage profile $\frac{1}{1-\alpha_M} \log Z_M(\ell) - \frac{1}{1-\alpha_A} \log Z_A(\ell) - \frac{\alpha_M}{1-\alpha_M} \log(1 + \psi_M(\ell))$ implies that $R_A(w, \ell) - R_M(w, \ell)$ crosses zero transversally at a unique interior boundary $b_{ct} \in (0, 1)$, assigning $[0, b_{ct}]$ to agriculture and $(b_{ct}, 1]$ to manufacturing. At the winning plan, local employment, worker participation, output, and wedge transfers satisfy

$$n(\ell) = \sum_{i \in \{A, M\}} e_i(\ell) L_i(\ell) + L_T(\ell), \tag{34}$$

$$\bar{u}_t = \mathbb{E}_t^\theta \left[\frac{w(\ell) + b_t - \gamma p_A(\ell)}{\mathcal{P}(p_A(\ell), p_M(\ell))} \right], \tag{35}$$

and realized output, innovation expenditure, and wedge transfers satisfy

$$Y_i = e_i Z_i G_i L_i^{\alpha_i}, \qquad \Psi_i = \Psi_i(\phi_i^*(\ell); w(\ell), \ell), \qquad \Pi_\psi = \sum_i e_i \psi_i w L_i. \tag{36}$$

PROPOSITION 2: *Fix the inherited physical state, dated environment, and a common rule for selecting conditional private equilibria. Conditional on an investment profile, current household and firm decisions and market prices are invariant to θ. If two belief specifications select the same profile, the same private discoveries and sectoral shocks generate the same current allocation and next physical state.* [Proof →]

Because $\mathbb{E}_t^\theta$ in (10) reweights only future aggregate shocks $\mathbf{z}_{t+1}$ while leaving the conditional distribution of date-t private discoveries $G_{it}(\ell, \omega)$ unchanged, every within-period expectation conditional on $(\Omega_t, \mathbf{x}_t)$ reduces to rational expectations ($\mathbb{E}_t^\theta[\cdot] = \mathbb{E}_t[\cdot]$ in (17), (27), and (35)); diagnostic expectations affect the allocation solely through county infrastructure investment $\mathbf{x}_t^\theta$.[6]

After early-stage plans are committed, the active producer's realized discovery $G_{it}(\ell) \geq 1$ enters the public knowledge pool, updating local pre-diffusion technology (Cai et al., 2022):

$$A_{it}^+(\ell) = A_{it}(\ell)\big[1 + e_{it}(\ell)(G_{it}(\ell) - 1)\big]. \tag{37}$$

The inactive sector ($e_{it}(\ell) = 0$) retains $A_{it}(\ell)$ at this step, and transitory aggregate gaps $\mathbf{z}_t$ do not enter permanent knowledge. Because spillovers attenuate exponentially with distance (Berkes et al., 2025; Keller, 2002), the technology frontier available at ℓ is the pointwise spatial supremum

$$F_{it}(\ell) = \sup_{r \in [0,1]} e^{-\delta|\ell - r|} A_{it}^+(r), \qquad \delta > 0, \tag{38}$$

which is uniquely defined everywhere under pointwise entrant optimality and land sorting, and finite almost surely because the discovery fields in (23) have continuous sample paths. Both sectors then close a fraction of their gap to $F_{it}(\ell)$ via the Nelson and Phelps (1966) catch-up law at a speed governed by installed route capacity (Berkes et al., 2026; Cinnirella et al., 2024):

$$A_{i,t+1}(\ell) = A_{it}^+(\ell) + (1 - e^{-\lambda_t(\ell)})[F_{it}(\ell) - A_{it}^+(\ell)], \tag{39}$$

$$\lambda_t(\ell) = \lambda_0 \left[\frac{T_t(\ell)}{T_{t_0}(\ell)}\right]^{1-\alpha_T}, \qquad \lambda_0 > 0, \tag{40}$$

where $T_{t_0}(\ell)$ is the initial reference network (half-life $\log(2)/\lambda_0$ years).

PROPOSITION 3: *For bounded positive post-discovery knowledge and finite nonnegative adoption rates, $A_{it}^+(\ell) \leq A_{i,t+1}(\ell) \leq F_{it}(\ell)$. Holding discoveries and reference levels fixed, more inherited capacity weakly raises next-period productivity. If no further discoveries occur and geography is fixed, the available frontier remains unchanged and*

$$F_{it}(\ell) - A_{i,t+h}(\ell) = [F_{it}(\ell) - A_{it}^+(\ell)] \exp\left[-\sum_{k=t}^{t+h-1} \lambda_k(\ell)\right], \qquad h \geq 1. \tag{41}$$

[Proof →]

Localized research in industrial hubs pushes out $F_{it}(\ell)$, while county transport investments raise $\lambda_t(\ell)$ and propagate discoveries across space. Because neither private entrants nor county promoters internalize how $T_t(\ell)$ accelerates future technology absorption, (39)–(40) generate a dynamic spatial adoption externality. Online Appendix S.C.2 proves that the frontier is finite and derives its moments and the exact conditional mean of next-period knowledge.

[6] Keeping within-period Bernoulli–Pareto discovery probabilities physical isolates the intertemporal network-investment channel, where construction must be committed a full period before operating returns realize.

E Transport and Distribution

Moving goods over land requires both durable route infrastructure (Donaldson, 2018; Redding and Turner, 2015) and operating carriage services priced in local transport markets (Brancaccio et al., 2020). Transport firms combine installed route capacity $T(\ell,t)$ with hired labor $L_T(\ell,t)$ committed at the local contract wage $w(\ell,t)$ before private discoveries, selling operating services ex post at the county toll τ_j:

$$Y_T(\ell,t) = Z_T(t) L_T(\ell,t)^{\alpha_T} T(\ell,t)^{1-\alpha_T}, \qquad Z_T(t) > 0, \quad 0 < \alpha_T < 1. \tag{42}$$

Each county $\mathcal{C}_j$ operates a competitive operating-service market at a common toll $\tau_{j(\ell)}$: services produced anywhere in $\mathcal{C}_j$ meet distribution or transit demand anywhere within $\mathcal{C}_j$ without additional cost, whereas services cannot be dispatched across county boundaries.

Shipping good i across space incurs both physical transit degradation at rate $\kappa(\ell,t)$ and r_i units of county operating services per unit of distance at toll $\tau_{j(\ell)}(t)$, yielding the effective ad valorem transport-cost rate $\tilde{\kappa}_i(\ell,t)$ and spatial no-arbitrage law for signed trade flow $H_i(\ell,t)$ (positive rightward):

$$\tilde{\kappa}_i(\ell,t) = \kappa(\ell,t) + \frac{r_i \tau_{j(\ell)}(t)}{p_i(\ell,t)}, \tag{43}$$

$$|\partial_\ell p_i| \le \kappa p_i + r_i \tau_{j(\ell)}, \qquad \partial_\ell p_i = (\kappa p_i + r_i \tau_{j(\ell)}) \operatorname{sign}(H_i) \quad \text{if } H_i \neq 0. \tag{44}$$

Because physical transit loss is proportional to p_i while carriage is paid per unit of service, spot prices $p_i(\cdot,t)$ are absolutely continuous across space: a step in $\tau_{j(\ell)}$ across a county border kinks $\partial_\ell p_i$ without creating a discontinuous price jump. Installed infrastructure lowers physical transit losses and bilateral iceberg delivery factors $d_t(\ell,r)$ exponentially:

$$\kappa(\ell,t) = \kappa_0 e^{-T(\ell,t)}, \qquad \kappa_0 > 0, \quad T(\ell,t) > 0, \tag{45}$$

$$d_t(\ell,r) = \exp\left\{ \kappa_0 \left| \int_r^\ell e^{-T_t(u)}\, du \right| \right\}. \tag{46}$$

Each unit of marketed output $Y_i(\ell,t)$ requires r_i units of county service for local origin distribution and r_i units per unit of distance shipped ($r_M = 1$, $r_A > 0$), so county-wide service demand and operating capacity are

$$D_j(\omega) = \int_{\mathcal{C}_j} \sum_{i \in \{A,M\}} r_i \big(Y_i(\ell,\omega) + |H_i(\ell,\omega)| \big)\, d\ell, \qquad K_j = \int_{\mathcal{C}_j} Y_T(\ell,t)\, d\ell. \tag{47}$$

Carrier optimization and county service clearing.— Before private discoveries, a competitive carrier at $\ell \in \mathcal{C}_j$ chooses $L_T(\ell)$ taking $T(\ell)$, $w(\ell)$, and the distribution of τ_j as given:

$$\max_{L_T \ge 0} \ \mathbb{E}_t^\theta[\tau_j\, Y_T - w L_T], \qquad Y_T = Z_T L_T^{\alpha_T} T^{1-\alpha_T}, \tag{48}$$

$$w = \alpha_T Z_T L_T^{\alpha_T - 1} T^{1-\alpha_T}\, \mathbb{E}_t^\theta[\tau_j] \quad \Longleftrightarrow \quad L_T^*(w,\ell) = \left[\frac{\alpha_T Z_T\, \mathbb{E}_t^\theta[\tau_{j(\ell)}]}{w(\ell)} \right]^{\frac{1}{1-\alpha_T}} T(\ell), \tag{49}$$

where $\mathbb{E}_t^\theta[\tau_j] = \mathbb{E}_t[\tau_j]$ conditional on $(\Omega_t, \mathbf{x}_t)$ by Proposition 2. After private discoveries realize, predetermined county capacity K_j clears against county service demand D_j via complementary slackness, and carrier operating surplus accrues to the household portfolio:

$$D_j \le K_j, \qquad \tau_j \ge 0, \qquad \tau_j (K_j - D_j) = 0, \tag{50}$$

$$\Pi_T(\ell,t) = \tau_{j(\ell)}(t) Y_T(\ell,t) - w(\ell,t) L_T(\ell,t). \tag{51}$$

F County Infrastructure Companies

Unlike centralized national network planners (Allen and Arkolakis, 2022; Fajgelbaum and Schaal, 2020), historical British turnpikes, canals, and railways were promoted by local county companies through parliamentary acts (Bogart, 2014). Each county $j \in \{1, \dots, J\}$ has an infrastructure company holding a long-lived claim on land-lease rents inside $\mathcal{C}_j$. Roads, waterways, and railways accumulate into a composite route stock $T(\ell, t)$ under one-period time-to-build (Kydland and Prescott, 1982): before private discoveries at date t, company j commits completion intensity $x_{jt} \in [0, \bar{x}_{jt}]$ within the resource-feasible statutory box $\mathcal{X}_t = \prod_j [0, \bar{x}_{jt}] \subseteq \prod_j [0, o_{jt}]$ ($o_{jt} \in \{0, 1\}$), and completed increments enter operating route capacity at date $t+1$:

$$x_{jt} \in [0, \bar{x}_{jt}], \qquad \operatorname{supp} \iota_{jt} \subseteq \mathcal{C}_j, \tag{52}$$

$$T_{t+1}(\ell) = T_t(\ell) + \sum_{j=1}^{J} x_{jt} \iota_{jt}(\ell), \qquad T_0(\ell) \geq \underline{T} > 0. \tag{53}$$

Construction uses manufactured goods with quadratic adjustment costs on the county engineering profile (e_{jt}, k_{jt}) ($\operatorname{supp} k_{jt} \subseteq \mathcal{C}_j$, $e_{jt} > 0$):

$$I_t(\ell; \mathbf{x}_t) = \sum_{j=1}^{J} m_I e_{jt} k_{jt}(\ell) \left(x_{jt} + \frac{\chi_I}{2} x_{jt}^2 \right), \qquad m_I > 0, \quad \chi_I > 0, \tag{54}$$

purchased at the realized spot price $p_M(\ell, t, \omega)$. Full-package physical increments grow at two percent per year, $\iota_{jt} = 1.02^{t-t_0} \iota_{j,t_0}$; after the last observed schedule year $t = 1881$, each county repeats its 1881 opportunity $o_{j,1881}$ and engineering profile $(e_j k_j)_{1881}$ at the same (m_I, χ_I) while physical increments continue to grow as $\iota_{j,1881+h} = 1.02^h \iota_{j,1881}$.

Promoter optimization, marginal-q, and stage Nash equilibrium.— Taking rival counties' intensities $\mathbf{x}_{-j,t}$ as given, company j's current payoff is expected county land rent net of spot manufacturing construction expenditure:

$$R_{jt}(\mathbf{x}; \Omega_t) = \mathbb{E}_t^\theta \left[\int_{\mathcal{C}_j} \{ R_t(\ell; \mathbf{x}, \Omega_t) - p_{Mt}(\ell, \omega; \mathbf{x}, \Omega_t) I_t(\ell; \mathbf{x}) \} \, d\ell \right]. \tag{55}$$

By Proposition 2, current returns conditional on $(\Omega_t, \mathbf{x})$ depend only on physical private discoveries ($\mathbb{E}_t^\theta = \mathbb{E}_t$ in (55)), whereas future aggregate shocks enter the continuation value under $\mathbb{E}_t^\theta$ with perceived next-period reference state $\overline{\mathbf{z}}_{t+1}^{\text{forecast}} = M(\mathbf{m}_t^\theta)$ from (12) (while the realized physical state at date $t+1$ resets to $\overline{\mathbf{z}}_{t+1} = M(\boldsymbol{\rho} \mathbf{z}_t)$):

$$\Pi_{jt}^\theta(\mathbf{x}; \Omega_t) = R_{jt}(\mathbf{x}; \Omega_t) + \beta \mathbb{E}_t^\theta \left[V_{j,t+1}^\theta(\Omega_{t+1}) \right], \qquad V_{jt}^\theta(\Omega_t) = \Pi_{jt}^\theta \left(\mathbf{x}_t^\theta(\Omega_t); \Omega_t \right), \tag{56}$$

$$x_{jt}^\theta(\Omega_t) \in \arg \max_{0 \leq u \leq \bar{x}_{jt}} \Pi_{jt}^\theta(u, \mathbf{x}_{-j,t}^\theta(\Omega_t); \Omega_t), \qquad j = 1, \dots, J. \tag{57}$$

If the pure-strategy Nash set $\mathcal{N}_t^\theta$ contains multiple profiles, parliamentary coordination selects the profile maximizing $\sum_{j=1}^J \Pi_{jt}^\theta$ (with deterministic tie-breaking). Because the aggregate spatial knowledge frontier and future knowledge trajectories $(A_{As}, A_{Ms})_{s \geq t+1}$ reflect atomistic firm discoveries and frontier adoption across the national continuum, each county company takes the aggregate knowledge path as external to its own corridor completion choice x_{jt} when evaluating continuation values. Where differentiable, county j's perceived marginal continuation value q_{jt}^θ (differentiating

with respect to own route capacity T_{t+1} at the given knowledge path) and marginal construction cost c_{jt} satisfy (Hayashi, 1982):

$$q^{\theta}_{jt}(\mathbf{x}) = \beta \partial_{x_j} \mathbb{E}^{\theta}_t[V^{\theta}_{j,t+1}(\Omega_{t+1})], \qquad c_{jt}(\mathbf{x}) = -\partial_{x_j} R_{jt}(\mathbf{x}; \Omega_t), \tag{58}$$

$$q^{\theta}_{jt}(\mathbf{x}^{\theta}_t) = c_{jt}(\mathbf{x}^{\theta}_t) \tag{59}$$

at an interior best response $x^{\theta}_{jt} \in (0, \bar{x}_{jt})$, with complementary slackness at $\{0, \bar{x}_{jt}\}$. Because every engineering profile lies inside its own county ($\operatorname{supp} k_{kt} \subseteq \mathcal{C}_k$), county j pays only for its own construction, and (55) splits into gross county land rent and a construction bill, $R_{jt}(\mathbf{x}; \Omega_t) = \widetilde{R}_{jt}(\mathbf{x}) - m_I(x_{jt} + \frac{\chi_I}{2} x^2_{jt}) D^I_{jt}(\mathbf{x})$, with

$$\widetilde{R}_{jt}(\mathbf{x}) = \int_{\mathcal{C}_j} R_t(\ell; \mathbf{x}, \Omega_t)\, d\ell, \qquad D^I_{jt}(\mathbf{x}) = \int_{\mathcal{C}_j} \mathbb{E}_t[p_{Mt}(\ell, \omega; \mathbf{x}, \Omega_t)]\, e_{jt} k_{jt}(\ell)\, d\ell > 0, \tag{60}$$

where D^I_{jt} is the county construction-price index. Differentiating gives the marginal construction cost c_{jt} and, collecting the marginal benefits of capacity, the gross marginal value $\widetilde{q}^{\theta}_{jt}$:

$$c_{jt} = m_I(1 + \chi_I x_{jt}) D^I_{jt} - \partial_{x_j} \widetilde{R}_{jt} + m_I\Big(x_{jt} + \frac{\chi_I}{2} x^2_{jt}\Big) \partial_{x_j} D^I_{jt},$$

$$\widetilde{q}^{\theta}_{jt} = q^{\theta}_{jt} + \partial_{x_j} \widetilde{R}_{jt} - m_I\Big(x_{jt} + \frac{\chi_I}{2} x^2_{jt}\Big) \partial_{x_j} D^I_{jt}.$$

Condition (59) therefore reads $\widetilde{q}^{\theta}_{jt} = m_I(1+\chi_I x_{jt}) D^I_{jt}$, and with complementary slackness it becomes the clipped investment rule

$$x^{\theta}_{jt} = \min\left\{\bar{x}_{jt},\ \max\left\{0,\ \frac{1}{\chi_I}\left[\frac{\widetilde{q}^{\theta}_{jt}(\mathbf{x}^{\theta}_t)}{m_I D^I_{jt}(\mathbf{x}^{\theta}_t)} - 1\right]\right\}\right\}. \tag{61}$$

A county builds only when the marginal value of capacity per unit of construction outlay exceeds one, and lower adjustment convexity χ_I makes buildout more responsive to that ratio. Diagonal dominance of own adjustment-cost curvature over cross-county general-equilibrium spillovers yields a unique stage equilibrium (Vives, 1999):

PROPOSITION 4: *Fix $(\Omega_t, \mathcal{E}_t)$ and $\theta \geq 0$ on a nonempty compact box $\mathcal{X}_t = \prod_j [0, \bar{x}_{jt}]$ where every profile has a unique admissible private allocation and continuous county payoffs Π^{θ}_{jt}. If each Π^{θ}_{jt} is concave in x_{jt}, $\mathcal{N}^{\theta}_t$ is nonempty and compact. If, in addition, Π^{θ}_{jt} is C^2 and there exist $m_j > 0$, $L_{jk} \geq 0$, and $v_j > 0$ on $\mathcal{X}_t$ such that*

$$-\partial_{jj}\Pi^{\theta}_j \geq m_j > 0, \qquad |\partial_{jk}\Pi^{\theta}_j| \leq L_{jk} \quad (k \neq j), \tag{62}$$

$$\gamma_I = \max_j \frac{\sum_{k \neq j} L_{jk} v_k}{m_j v_j} < 1, \tag{63}$$

then the county Nash equilibrium is unique and simultaneous best responses converge to it from every initial profile in $\mathcal{X}_t$. [Proof →]

By Proposition 2, θ enters the dated allocation solely through $q^{\theta}_{jt}(\mathbf{x})$ in (59). Differentiating the diagnostic kernel (10)–(11) holding $V^{\theta}_{j,t+1}$ fixed shows that current news $\boldsymbol{\varepsilon}_t = \mathbf{z}_t - \overline{\mathbf{z}}_t$ shifts perceived marginal-q directly and—together with $dq^{\theta}_{jt}/d\theta = \partial_{\theta} q^{\theta}_{jt}\big|_{V_{t+1}} + \beta \mathbb{E}^{\theta}_t[\partial_{x_j} \partial_{\theta} V^{\theta}_{j,t+1}]$—shifts interior buildout ($\mathcal{I}_t = \{j : x^{\theta}_{jt} \in (0, \bar{x}_{jt})\}$) according to

$$\begin{aligned} \partial_{\theta} q^{\theta}_{jt}(\mathbf{x})\Big|_{V_{t+1}} &= \beta \operatorname{Cov}^{\theta}_t\Big(\partial_{x_j} V^{\theta}_{j,t+1}(\Omega_{t+1}),\ \mathbf{z}'_{t+1} \boldsymbol{\Sigma}^{-1} \boldsymbol{\rho} \boldsymbol{\varepsilon}_t\Big), \\ \frac{d\mathbf{x}^{\theta}_{t,\mathcal{I}_t}}{d\theta} &= -\left[D_{\mathbf{x}_{\mathcal{I}_t}}(\mathbf{q}^{\theta}_{t,\mathcal{I}_t} - \mathbf{c}_{t,\mathcal{I}_t})\right]^{-1} \frac{d\mathbf{q}^{\theta}_{t,\mathcal{I}_t}(\mathbf{x}^{\theta}_t)}{d\theta}. \end{aligned} \tag{64}$$

Because each county company captures only local land rents inside $\mathcal{C}_j$ and takes aggregate knowledge evolution as external, its private marginal-q omits three positive social returns to route capacity: (i) cross-county market-access and transit savings $\sum_{k\neq j}\partial_{x_j}\Pi^0_{kt} > 0$ (Allen and Arkolakis, 2022; Donaldson and Hornbeck, 2016; Fajgelbaum and Schaal, 2020; Fogel, 1964; Krugman, 1991); (ii) the allocative-efficiency gain $\int_0^1 \psi_M(\ell)w(\ell)\partial_{x_j}L_{Mt}(\ell)\,d\ell > 0$ from reallocating labor into wedge-distorted manufacturing (Baqaee and Farhi, 2020; Hornbeck and Rotemberg, 2024; Hsieh and Klenow, 2009; Restuccia and Rogerson, 2008); and (iii) faster spatial knowledge diffusion (Aghion and Howitt, 1992; Desmet and Rossi-Hansberg, 2014; Lucas, 1988; Mokyr, 2005; Nelson and Phelps, 1966; Romer, 1990), since higher $T_s(\ell)$ at $s \geq t+1$ raises next-period knowledge at given discoveries by

$$\frac{\partial A_{i,s+1}(\ell)}{\partial T_s(\ell)} = (1-\alpha_T)\frac{\lambda_s(\ell)}{T_s(\ell)}e^{-\lambda_s(\ell)}\big[F_{is}(\ell) - A^+_{is}(\ell)\big] \geq 0, \tag{65}$$

which is strictly positive wherever the available frontier exceeds local technology. Rational expectations ($\theta = 0$) therefore underinvest across all three margins, whereas favorable manufacturing and transport news raises $q^\theta_{jt} > q^0_{jt}$ and expands irreversible network buildout under diagnostic expectations ($\theta > 0$).

G Equilibrium and State Transitions

Fix $(\Omega_t, \mathcal{E}_t)$ and a feasible county investment profile $\mathbf{x}_t \in \mathcal{X}_t$. Before private discoveries, pre-discovery plans $(e_i, \phi_i, L_i, L_T, n, w)$ clear local and national labor markets, while ex-post goods flows $H_i(\ell, t)$ satisfy spatial conservation with closed borders:

$$n = \sum_{i\in\{A,M\}} e_i L_i + L_T, \tag{66}$$

$$\int_0^1 n\,d\ell = \bar{L}_t, \tag{67}$$

$$\partial_\ell H_i = Y_i - nc_i - e_i\frac{\Psi_i}{p_i} - I_i - \kappa|H_i|, \qquad H_i(0,t) = H_i(1,t) = 0, \quad i \in \{A, M\}. \tag{68}$$

Realized producer net profit Π_F, contractual land rent R, origin carrier surplus Π_T, manufacturing wedge rebates Π_ψ, net land rent R^{net}, and the common portfolio dividend b_t are

$$\Pi_F = \sum_i e_i\Big\{(p_i - r_i\tau_{j(\ell)})Y_i - (1+\psi_i)wL_i - \Psi_i(\phi_i; w, \ell)\Big\} - R, \tag{69}$$

$$R = \sum_i e_i R_i, \qquad \Pi_T = \tau_{j(\ell)}Y_T - wL_T, \qquad \Pi_\psi = \sum_i e_i\psi_i wL_i, \tag{70}$$

$$R^{\text{net}} = R - p_M I, \tag{71}$$

$$b_t = \frac{1}{\bar{L}_t}\int_0^1 \{R^{\text{net}} + \Pi_F + \Pi_T + \Pi_\psi\}\,d\ell. \tag{72}$$

Free entry implies $\mathbb{E}_t[\Pi_F] = 0$, while ex-post windfalls $\Pi_F(\ell, \omega)$ accrue to households. Carrier tolls and wedge rebates are pecuniary transfers, whereas research goods Ψ_i/p_i, construction I, and iceberg degradation $\kappa|H_i|$ absorb physical output.

DEFINITION 1 (Conditional dated equilibrium): Given $(\Omega_t, \mathcal{E}_t, \mathbf{x}_t)$, a conditional dated equilibrium consists of pre-discovery plans $(e_i, \phi_i, L_i, L_T, n, w, R, I)$ and a measurable post-discovery family $(Y_i, c_i, p_i, \tau_j, H_i, b_t)$ such that: (i) entrants solve (28), land is assigned by (33), and utilization solves (26); (ii) households choose residence by (16) and consumption by (15); (iii) carriers solve

(48) and county services clear by (50); and (iv) for almost every realization, spatial no-arbitrage, goods and labor clearing, and (72) hold. At continuous versions of $p_i(\cdot,\omega)$ and $w(\cdot)$ on $[0,1]$ (with $p_A(1,t)=1$), entrant optimality (28) and pure-site sorting (33) hold at every $\ell\in[0,1]$ (breaking isolated ties to the left sector), differential flow and no-arbitrage relations hold almost everywhere, subsistence is feasible statewise, and expectations in agent objectives are finite.

In a competitive continuum economy (Aumann, 1966), individual entrants and carriers evaluate alternative plans at the quoted price and contract-wage functions (w,p,τ) without recomputing local market clearing, while equilibrium announcements jointly clear labor, goods, and county service markets.

PROPOSITION 5: *For every realization satisfying local labor clearing, spatial goods balance, spatial no-arbitrage, and closed national boundaries, the ledgers (69)–(72) imply the aggregate household budget identity. By Walras' Law, there is no additional independent national resource constraint.* [Proof →]

Building on classical general equilibrium analysis (Arrow and Hahn, 1971) and the nonlinear spatial operator representation of Allen and Arkolakis (2014) and Allen et al. (2024), substituting the closed-form household demand and mobility rules (15)–(17), producer hiring and research rules (29)–(31), land sorting (33), and carrier hiring (49) into market clearing (66)–(72) reduces the conditional dated equilibrium to a fixed point $(w,p,\tau)=\mathcal{T}_t(w,p,\tau)$ on the quote domain $\mathcal{W}_t\times\mathcal{P}_t\times\mathcal{M}_t$ of positive continuous wages $w\in C([0,1];[\underline{w}_t,\overline{W}_t])$, statewise continuous spot prices $p=(p_A,p_M)$ $(p_A(1,\omega)=1)$, and nonnegative county tolls $\tau=(\tau_1,\dots,\tau_J)$ with positive expected tolls $\mathbb{E}_t[\tau_j]\in[\underline{\tau}_{jt},\overline{\tau}_{jt}]$ $(\underline{\tau}_{jt}>0)$:

$$\begin{aligned}\widetilde{w}(\ell)=\big(\mathcal{T}_t^{w}(w,p,\tau)\big)(\ell)&=\frac{\bar{u}_t(w,p,\tau)-\mathbb{E}_t\big[(b_t(\omega;w,p,\tau)-\gamma p_A(\ell,\omega))/\mathcal{P}(\ell,\omega)\big]}{\mathbb{E}_t\big[1/\mathcal{P}(\ell,\omega)\big]},\\ \big(\widetilde{p}(\cdot,\omega),\widetilde{\tau}(\omega)\big)&=\mathcal{T}_t^{p}(w,p,\tau;\omega),\\ \bar{L}_t&=\int_0^1\Big[\sum_{i\in\{A,M\}}e_i(\ell)L_i^*\big(\phi_i^*(\widetilde{w}(\ell),\ell);\widetilde{w}(\ell),\ell\big)+L_T^*(\widetilde{w}(\ell),\ell)\Big]d\ell,\end{aligned}\tag{73}$$

where national labor clearing at $\widetilde{w}=\mathcal{T}_t^{w}(w,p,\tau)$ pins down $\bar{u}_t(w,p,\tau)$ and $\mathcal{T}_t^{p}(w,p,\tau;\omega)$ maps predetermined capacities and construction demand $I_t(\cdot;\mathbf{x}_t)$ into the unique spot prices and county tolls supporting the strictly concave Negishi-weighted log-surplus spatial dispatch problem under (44), (50), and (68)–(72).

Existence and uniqueness of the conditional dated equilibrium rest on three primitive conditions:

1. *(Resource feasibility and finite innovation tails.)* On the contract-wage collar $[\underline{w}_t,\overline{W}_t]$ where zero-research labor demand crosses $\bar{L}_t$, net deliverable agricultural capacity exceeds subsistence $(\underline{Y}_{At}>\gamma\bar{L}_t)$, net deliverable manufacturing capacity exceeds construction demand $(\underline{Y}_{Mt}>\int_0^1 I_t(\ell;\mathbf{x}_t)\,d\ell)$, county carrier capacity covers distribution demand uniformly on $\mathcal{X}_t$, and $a_i>1/(1-\alpha_i)$ for $i\in\{A,M\}$.

2. *(Innovation-cost convexity and Ricardian single-crossing.)* $\chi_i>\bar{\chi}_{it}$ (Appendix A.6.1) so active research (30) is strictly concave on $(0,1)$, $\pi_i^0(w(\ell),\ell)-\underline{\pi}_i(\ell)$ crosses zero transversally at most finitely many times when $f_i>0$, and $\partial_\ell[\frac{1}{1-\alpha_M}\log Z_{Mt}(\ell)-\frac{1}{1-\alpha_A}\log Z_{At}(\ell)-\frac{\alpha_M}{1-\alpha_M}\log(1+\psi_M(\ell))]>0$ dominates spatial factor-price and toll gradients on $\mathcal{U}_t$ so the bid-rent difference $R_A(\ell)-R_M(\ell)$ satisfies $R_A(0)>R_M(0)$, $R_A(1)<R_M(1)$, and $\partial_\ell[R_A(b_{ct})-R_M(b_{ct})]<0$ at a unique cutoff $b_{ct}\in(0,1)$.

3. *(Norm contraction.)* In a weighted Banach norm $\|\cdot\|_{\mathcal{U}_t}$, the Jacobian $\mathcal{D}_t(u)$ of $\mathcal{T}_t$ satisfies $\sup_{u\in\mathcal{U}_t}\|\mathcal{D}_t(u)\|_{\mathcal{U}_t} \le \bar{\rho}_t < 1$, so direct local labor-demand elasticities $1/(1-\alpha_i)$ dominate indirect spatial trade, toll, and dividend feedbacks.

PROPOSITION 6: *Consider the economy of Definition 1 with $0 < \alpha_A < \alpha_M < 1$, $\alpha_T \in (0,1)$, bounded positive piecewise-continuous (A_{At}, A_{Mt}), $\eta \in (0,1)$, $\gamma > 0$, and $\phi_{it}(\ell) \in [0,1)$.*

1. (Existence, active-research threshold, and pure-site sorting.) *Under Conditions 1 and 2, every $\mathbf{x}_t \in \mathcal{X}_t$ admits a regular conditional dated equilibrium in $\mathcal{U}_t$. At the continuous price and wage versions, optimal research $\phi^*_{it}(\ell) \in [0,1)$ is uniquely determined at every $\ell \in [0,1]$ by the threshold rule (31)–(32), and land partitions at a unique cutoff $b_{ct} \in (0,1)$ with agriculture on $[0, b_{ct}]$ and manufacturing on $(b_{ct}, 1]$, uniquely determining $A^+_{it}(\ell,\omega)$ and the almost surely finite frontier $F_{it}(\ell,\omega)$ (Online Appendix Lemma S.C.1) everywhere on $[0,1]$.*

2. (Global uniqueness.) *Under Condition 3, the regular conditional dated equilibrium is globally unique for each $\mathbf{x}_t \in \mathcal{X}_t$.*

3. (Constructive verification from primitives.) *Around the closed-form frictionless spatial benchmark, backward county food-anchor induction and regular perturbation verify a stock-uniform wage collar $[\underline{w}_{\text{nat}}, \overline{W}_{\text{nat}}]$ and Conditions 1–3 directly from primitives across all admissible county infrastructure paths $T_t(\ell) = T_0(\ell) + s_{j(\ell),t}(\ell) \ge T_0(\ell)$ (Appendix A.6.3).* [Proof →]

DEFINITION 2 (Dynamic equilibrium): Given Ω_0, diagnosticity $\theta \ge 0$, $\{\mathcal{E}_t\}_{t\ge 0}$, and the physical shock laws, a dynamic equilibrium is an adapted process of states, investments, conditional dated equilibria, and county values such that: (i) each dated allocation satisfies Definition 1; (ii) county policies satisfy (57) under values (56) and perceived forecast-reference update (12), taking the aggregate knowledge trajectory (A_{At}, A_{Mt}) as given; and (iii) realized states follow (9), (37)–(39), and (53) with observed reference $\bar{\mathbf{z}}_{t+1} = M(\boldsymbol{\rho}\mathbf{z}_t)$.

Substituting the unique conditional dated equilibrium rent and manufacturing price schedules from Proposition 6 into (55) yields county j's reduced stage payoff

$$R_{jt}(\mathbf{x}_t;\Omega_t) = \widetilde{R}_{jt}(\mathbf{x}_t) - m_I\left(x_{jt} + \frac{\chi_I}{2}x_{jt}^2\right) D^I_{jt}(\mathbf{x}_t),$$

with gross county land rent $\widetilde{R}_{jt}$ and construction-price index D^I_{jt} from (60), county mean route stock $\overline{T}_{j,t+1} = \overline{T}_{jt} + d_{jt}x_{jt}$ ($\overline{T}_{jt} = |\mathcal{C}_j|^{-1}\int_{\mathcal{C}_j} T_t(\ell)\,d\ell$ and $d_{jt} = |\mathcal{C}_j|^{-1}\int_{\mathcal{C}_j}\iota_{jt}(\ell)\,d\ell$ by (53)) and knowledge transition (37)–(39). After 1881, household mass remains at $\bar{L}_{1881+h} = \bar{L}_{1881}$ while the 1881 project schedule repeats with physical package growth $g = 1.02$. Write $\bar{Z}_T \equiv A_T/\mathbb{E}_\pi[e^{z_T}]$, $\varepsilon = e^{-z_{Tt}/\alpha_T}/(\bar{Z}_T)^{1/\alpha_T}$ for the carrier-friction scale, and $D_{\min} > 0$ for the lower bound on the construction-price index D^I_{jt}.

PROPOSITION 7: *Suppose the conditions of Proposition 6 hold along feasible state histories.*

1. (Finite- and infinite-horizon Markov perfect equilibrium.) *Whenever own adjustment-cost convexity $m_I\chi_I D_{\min}$ dominates cross-county price and continuation spillovers, the J-county stage pseudogradient is strongly monotone on $\mathcal{X}_t$ with modulus $\mu = \frac{1}{8}m_I\chi_I D_{\min} > 0$, attenuating cross-county policy sensitivity to inherited county stock $\overline{T}_{it}$:*

$$\left\|\frac{\partial \mathbf{x}_t^{(H),\theta}}{\partial \overline{T}_{it}}\right\| \le \frac{C_u\varepsilon}{\mu \min_r \overline{T}_{rt}^{(1+\alpha_T)/\alpha_T}}. \tag{74}$$

For every finite horizon $H \geq 1$ and $\theta \in [0, \bar{\theta}]$, backward induction yields a unique pure-strategy Markov perfect equilibrium $(\mathbf{x}_t^{(H),\theta}, \mathbf{V}_t^{(H),\theta})_{t=0}^{H}$, which converges geometrically as $H \to \infty$ to a unique infinite-horizon dynamic equilibrium $(\mathbf{x}_t^{\theta}, \mathbf{V}_t^{\theta})_{t\geq 0}$ whenever the state-weight drift κ and policy sensitivity satisfy $\beta\kappa + BL_a < 1$ (Theorem A.1).

2. (Constructive verification on the rescaled corridor.) *Under the terminal schedule with $\beta g^{1-\alpha_T} < 1$, regular perturbation around the frictionless benchmark verifies stage-game strong monotonicity, horizon-invariant continuation concavity, and geometric convergence to a unique interior infinite-horizon equilibrium on a forward-invariant rescaled stock corridor (Appendix A.7).* [Proof →]

IV Quantitative Analysis

This section estimates the model on historical evidence from England and Wales over 1710–1881 and uses the estimated model to quantify the contribution of diagnostic expectations to transport accumulation, structural transformation, real wages, prices, and frontier knowledge during the industrial takeoff (Broadberry et al., 2014; Crafts and Harley, 1992). Spatial allocations are computed on $N = 500$ locations partitioned into $J = 10$ county jurisdictions, solving at each annual date for the conditional dated equilibrium and the Markov perfect county investment profile.

A From Historical Districts to Model Space

Because England and Wales is predominantly north–south oriented, latitude provides a natural one-dimensional ordering of regional economic activity on $\ell \in [0, 1]$ (Bogart et al., 2022; Trew, 2020). For each of twenty-five equal-latitude bands, historical employment and route quantities are summed across the band's east–west extent and divided by land width, with $\ell = 0$ at the southern coast and $\ell = 1$ at the northern border.

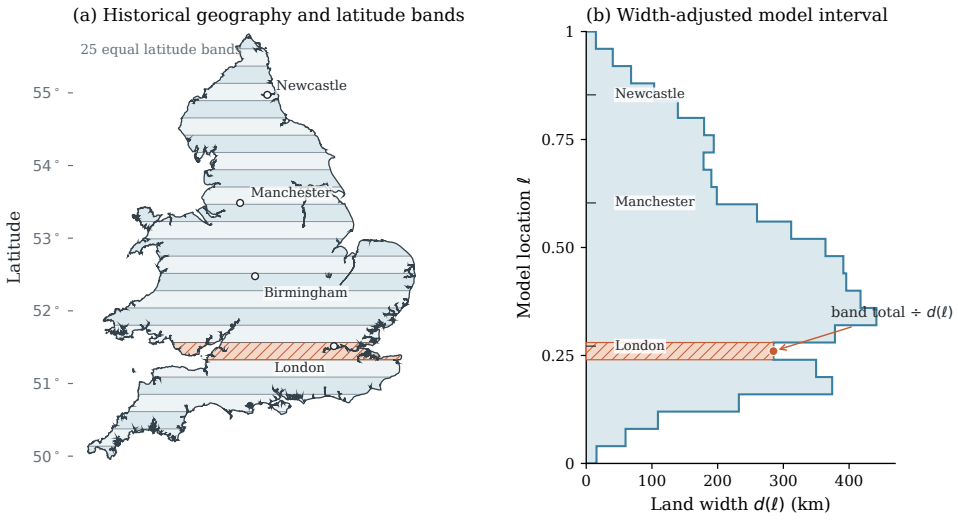


Figure 6: From Historical Geography to the Model Interval

Notes: Panel (a) partitions England and Wales into 25 equal-latitude bands. Panel (b) divides band totals by east–west land width $d(\ell)$ to form densities on $\ell \in [0, 1]$.

This projection captures the relative positions of metropolitan London in the South, the West

Midlands, and the northern coalfield belt, together with the reallocation of labor across them, while abstracting from east–west variation within a latitude band (Online Appendix S.B).

B Data and External Calibration

Aggregate household mass $\bar{L}_t$ follows the annual England and Wales population series of Thomas and Dimsdale (2017) over 1710–1881, normalized to $\bar{L}_{1710} = 100$ and held constant after 1881 ($\bar{L}_{1881+h} = \bar{L}_{1881}$). Installed transport capacity is reconstructed annually across the twenty-five latitude bands from turnpike, waterway, and railway GIS records. Letting M^r_{hy}, M^w_{hy}, and M^g_{hy} denote permanent route miles of turnpike road (Rosevear et al., 2026), navigable waterway (Cambridge Group for the History of Population and Social Structure, 2016a), and railway (Cambridge Group for the History of Population and Social Structure, 2016b; Coll Ardanuy et al., 2021; Mitchell, 1988) in band h and year y, and d_h the band's mean east–west land width, the measured transport stock is

$$\mathcal{I}^T_{hy} = \frac{(M^r_{hy} + 10M^w_{hy} + 50M^g_{hy})/d_h}{25^{-1}\sum_{k=1}^{25}(M^r_{k,1710} + 10M^w_{k,1710} + 50M^g_{k,1710})/d_k}, \tag{75}$$

where the service-capacity weights $(1, 10, 50)$ follow Trew (2020) and the 1710 cross-band mean is normalized to unity. The 1710 cross-section initializes $T_0(\ell)$, while subsequent stocks are generated endogenously by county investment (52)–(53).

Following Hornbeck and Rotemberg (2024), the manufacturing operating wedge $\psi_M(\ell) = \bar{\psi}_M \omega_M(\ell)$ combines a scale parameter $\bar{\psi}_M$ with a predetermined spatial exposure profile $\omega_M(\ell)$ from geocoded fire-insurance registers (Study 1838), normalized to unit 1710 manufacturing-labor-weighted mean ($\frac{\int_0^1 \omega_M(\ell)\omega_{M0}(\ell)\,d\ell}{\int_0^1 \omega_{M0}(\ell)\,d\ell} = 1$ with $\omega_{M0}(\ell) \equiv e_{M,1710}(\ell)L_{M,1710}(\ell)$). By the conditional hiring rule (29), an active manufacturing site with payroll $w(\ell)L_M(\ell)$ and expected gross operating surplus $\mathbb{E}_t[F_{M\ell}] = \mathbb{E}_t\big[[p_M(\ell) - r_M\tau_{j(\ell)}]_+ Y_M(\ell)\big] - w(\ell)L_M(\ell)$ satisfies

$$1 + \psi_M(\ell) = \alpha_M\left(1 + \frac{\mathbb{E}_t[F_{M\ell}]}{w(\ell)L_M(\ell)}\right), \qquad \text{or} \qquad \frac{\alpha_M}{1-\alpha_M}\frac{\mathbb{E}_t[F_{M\ell}]}{w(\ell)L_M(\ell)} = 1 + \frac{\psi_M(\ell)}{1-\alpha_M}, \tag{76}$$

directly calibrating $\bar{\psi}_M = 0.251$ (SE $= 0.032$) from the Clark (2002) nonagricultural labor share, with digitized firm operating accounts (Clough records) providing an external check on $\mathbb{E}_t[F_{M\ell}]/[w(\ell)L_M(\ell)]$.

C Estimation and Identification

The estimation strategy is the Simulated Method of Moments (Bordalo et al., 2026; Hansen, 1982), which jointly estimates $p = 26$ free structural parameters under diagnostic expectations ($\theta \geq 0$ plus the 16 preference, innovation, spatial diffusion, transport, and construction-cost parameters and the 9 sectoral shock parameters in Table 5), while calibrating the manufacturing wedge scale $\bar{\psi}_M = 0.251$ (SE $= 0.032$) from (76) and concentrating out the eight shape parameters of the five-knot initial 1710 productivity profiles $A_A(\ell, 1710)$ and $A_M(\ell, 1710)$ ($\int_0^1 A_i(\ell, 1710)\,d\ell = 1$) on the 1710 occupational cross-section. To evaluate belief extrapolation, the model is also estimated under the rational expectations restriction ($\theta = 0$, $p = 25$ free SMM parameters). Both specifications share the fixed parameters in Table 3 and minimize the weighted distance between the $M = 30$ historical moments and simulated moments averaged across B common primitive shock histories $\{u_b\}_{b=1}^B$:

$$Q(\Theta) = \big[\overline{\mathbf{m}}(\Theta) - \mathbf{m}^{\text{data}}\big]' W \big[\overline{\mathbf{m}}(\Theta) - \mathbf{m}^{\text{data}}\big], \qquad \overline{\mathbf{m}}(\Theta) = \frac{1}{B}\sum_{b=1}^{B}\mathbf{m}(\Theta; u_b), \tag{77}$$

where $\mathbf{m}^{\text{data}} \in \mathbb{R}^{30}$ stacks the aggregate, transport-network, spatial-covariance, sectoral-cycle, and investment-dynamics moments in Panels A–E of Table 6, and $W = \text{diag}(1/\widehat{\text{SE}}_1^2, \ldots, 1/\widehat{\text{SE}}_{30}^2)$ weights each residual by its inverse empirical variance so that $Q(\Theta) = \sum_{k=1}^{30}\left[(\overline{m}_k(\Theta) - m_k^{\text{data}})/\widehat{\text{SE}}_k\right]^2$ is measured in studentized units (Online Appendix S.D and Table S.D.2 report the formula, standard error $\widehat{\text{SE}}_k$, and weight W_{kk} for each moment).

Table 3: Externally Fixed Parameters and Normalizations

Parameter	Value	Description
β	1/1.05	Annual discount factor (real interest rate of 5%)
$\alpha_A,\ \alpha_M$	0.390, 0.710	Production labor elasticities (agriculture, manufacturing)
r_M	1.000	Manufacturing distribution requirement (normalization)
c	3	Truncation bound for aggregate productivity innovations (in s.d.)
d_0	0.100	Spatial Ornstein–Uhlenbeck correlation length for discovery draws
g_m	log(1.02)	Post-1881 growth rate of physical project increments
$\bar{L}_{1710}$	100	Initial aggregate labor supply (normalization)
$N,\ J$	500, 10	Number of spatial grid cells and county jurisdictions

Notes: Numeraire $p_A(1,t) = 1$; initial productivity profiles satisfy $\int_0^1 A_{i,1710}(\ell)\,d\ell = 1$ $(i \in \{A, M\})$; and the 1710 transport stock has cross-band mean one. Production labor elasticities (α_A, α_M) follow sectoral value-added shares (Herrendorf et al., 2015) (Proposition S.E.2), and demographic mass $\bar{L}_t$ follows (8).

Table 4 summarizes how the moments discipline each parameter block. By Proposition 2, the allocation at each date depends on θ only through county construction, so within-period margins discipline the static parameters given the network, while network dynamics discipline beliefs and construction costs. Sectoral labor shares map into the fixed production elasticities (α_A, α_M), and the nonagricultural labor share, given fire-insurance exposure, calibrates $\bar{\psi}_M$ (76). The 1710 occupational shares, the relative price path, and dated spatial employment covariances discipline (η, γ), $(\alpha_T, \kappa_0, r_A)$, and $A_i(\ell, 1710)$. Aggregate output growth and the shift from agricultural to industrial output identify the innovation and diffusion parameters $(a_i, f_i, \chi_i, \omega_E, \delta, \lambda_0)$, while HP-filtered sectoral output variances, autocovariances, and cross-covariances pin down $(\boldsymbol{\rho}, \boldsymbol{\Sigma})$. Finally, the linear and quadratic trends and subperiod growth rates of log transport capacity (Panel B of Table 6) and the investment-cycle regressions (Panel E) separate θ from $(m_I, \chi_I, \boldsymbol{\rho})$. Under $\theta = 0$, lowering χ_I or raising $\boldsymbol{\rho}$ increases investment–output comovement only by raising the first-order autocorrelation ρ_1 of cyclical investment, whereas $\theta > 0$ amplifies boom-time marginal q, raises 1830–1850 transport growth, and lowers ρ_1 as transitory news subsides. These sign restrictions are verified numerically at the estimates.[7]

Following Bordalo et al. (2026), Table 5 reports the parameter estimates and standard errors under diagnostic (DE) and rational (RE, $\theta = 0$) expectations, and Table 6 compares the empirical and simulated moments across all five blocks. Diagnosticity is estimated at $\widehat{\theta} = 0.739$ (SE = 0.090), rejecting $\theta = 0$ ($t = 8.21$, $p < 0.0001$). This estimate lies within the benchmark range across financial and macroeconomic settings (Bordalo et al., 2026): slightly below the $\theta \approx 0.8$–0.9 estimated from earnings expectations and credit spreads (Bordalo et al., 2018, 2019), above the $\theta \approx 0.5$–0.6 from macroeconomic surveys (Bordalo et al., 2020), and aligned with the $\theta \approx 0.74$–1.08 in structural models of investment and credit cycles (Bianchi et al., 2024; Bordalo et al., 2026; Maxted, 2023). When the remaining $p = 25$ parameters are re-estimated under $\theta = 0$, the restricted RE estimator lowers adjustment convexity ($\widehat{\chi}_I = 1.579$ versus 1.920) and construction scale ($\widehat{m}_I = 0.020$

[7] Online Appendix S.E states the local rank conditions (Andrews et al., 2017; Rothenberg, 1971) and sign restrictions; Replication Appendix R.A details the sequence-space Jacobian evaluation of (77) (Auclert et al., 2021).

Table 4: Parameter Blocks and Identifying Moments

Parameter block	Parameters	Primary identifying moments
Preferences	$\eta,\ \gamma$	Linear trends in sectoral employment shares and log relative price $\ln(p_M/p_A)$
Operating wedge	$\bar{\psi}_M$	Nonagricultural labor share and fire-insurance exposure (76)
Innovation	$a_i,\ f_i,\ \chi_i,\ \omega_E$	Linear trends in log real GDP and $\ln(Y_M/Y_A)$; manufacturing share–energy covariance $C_{e,y}$
Spatial diffusion	$\delta,\ \lambda_0$	Manufacturing share–latitude covariance $C_{l,y}$ and spatial autocovariance $C_{N,y}$
Transport	$\alpha_T,\ \kappa_0,\ r_A$	Tertiary employment share trend, cross-band variance $V^{\text{occ}}_{T,y}$, and $\ln(p_M/p_A)$ trend
Construction costs	$m_I,\ \chi_I$	Linear and quadratic trends of log transport stock $\ln \overline{T}_t$; cross-band variance $V_{T,t}$
Diagnostic beliefs	θ	Subperiod transport growth (1830–1850), post-1850 slope change, investment–output slopes (b_0, b_1), autocorrelation (ρ_1), and county lag slopes (g_1, g_2)
Initial productivity	$A_i(\ell, 1710)$	1710 occupational shares and cross-sectional employment covariances
Productivity process	$\boldsymbol{\rho},\ \boldsymbol{\Sigma}$	Variances, first-order autocovariances, and cross-covariances of HP-filtered log sectoral output

Notes: Primary identifying moments for each parameter block (Table 6 and Online Appendix S.D); all blocks are estimated jointly via (77) under DE ($\theta \geq 0$) and RE ($\theta = 0$). Online Appendix S.E states the local rank and sign restrictions.

versus 0.023), raises manufacturing persistence ($\widehat{\rho}_M = 0.561$ versus 0.533) and volatility ($\widehat{\sigma}_M = 0.042$ versus 0.039), and adjusts spatial catch-up and transport costs ($\widehat{\lambda}_0 = 0.175$ versus 0.160, $\widehat{\kappa}_0 = 0.073$ versus 0.080, $\widehat{\delta} = 1.245$ versus 1.350) to compensate for the absence of extrapolation.

Because those parameter shifts distort other margins, the fit improvement under DE spans Table 6. Both models match long-run output and employment trends (Panel A) and dated spatial sorting covariances (Panel C). The restricted RE model, however, underpredicts the linear and quadratic trends of the log transport stock (1.758 versus 1.948 percent per year and 2.144 versus 2.923 in Panel B), the 1830–1850 railway boom (4.801 versus 7.410 percent per year under DE and 7.470 in the data), and the post-1850 slowdown (−2.461 versus −5.060 and −5.101). Raising $\widehat{\rho}_M$ and $\widehat{\sigma}_M$ under RE also overstates manufacturing and transport cycle variances ($v_M = 12.12$ versus 9.96 under DE and 9.64 in the data; $v_T = 7.37$ versus 5.86 and 5.66 in Panel D), shifts investment comovement from current to lagged output ($b_0 = 2.063$ and $b_1 = 3.142$ under RE, versus 3.091 and 1.662 under DE and 2.954 and 1.585 in the data), and mutes both the negative serial correlation of investment ($\rho_1 = -0.183$ versus −0.344 under DE and −0.325 in the data) and county investment reversals after population growth ($g_1 = -0.082$ and $g_2 = -0.144$ under RE, versus −0.190 and −0.484 under DE and −0.162 and −0.496 in the data; Panel E). Online Appendix Table S.E.1 confirms that this fit gain does not stem from an extra parameter or moment weighting: the Rivers and Vuong (2002) equal-parameter test ($p = 25$ free parameters in every column) favors $\theta = 0.9$ ($Q = 5.54$) over $\theta = 0$ ($Q = 117.86$) with $z = 4.35$ ($p < 0.0001$). The Andrews et al. (2017) sensitivity measures show that Panels E and B supply 85.7 percent (49.8 and 35.9 percent) of the identifying variation for $\widehat{\theta}$, while leave-one-block-out estimations yield $\widehat{\theta} \in [0.666, 0.747]$ and equal-block studentized weights yield $\widehat{\theta}_{\text{eq}} = 0.748$ (SE = 0.104).

Table 5: SMM Parameter Estimates: Diagnostic versus Rational Expectations

Parameter	Description	DE	RE ($\theta = 0$)
Panel A. Preferences, production, and trade costs			
η	Stone–Geary food expenditure share	0.520 (0.028)	0.540 (0.031)
γ	Subsistence food requirement	0.042 (0.003)	0.039 (0.003)
α_A	Agricultural labor elasticity	0.390 [fixed]	0.390 [fixed]
α_M	Manufacturing labor elasticity	0.710 [fixed]	0.710 [fixed]
$\bar{\psi}_M$	Manufacturing labor wedge scale	0.251 (0.032) [calibrated]	0.251 (0.032) [calibrated]
α_T	Transport labor elasticity	0.450 (0.029)	0.473 (0.033)
κ_0	Iceberg transport cost scale	0.080 (0.007)	0.073 (0.007)
r_A	Agricultural distribution requirement ($r_M = 1$)	0.200 (0.015)	0.216 (0.018)
Panel B. Innovation and spatial diffusion			
a_A	Agricultural innovation Pareto shape	2.200 (0.150)	2.122 (0.156)
a_M	Manufacturing innovation Pareto shape	1.800 (0.112)	1.887 (0.126)
f_A	Agricultural fixed innovation cost	0.040 (0.004)	0.043 (0.005)
f_M	Manufacturing fixed innovation cost	0.030 (0.003)	0.028 (0.003)
χ_A	Agricultural variable innovation cost	0.080 (0.008)	0.085 (0.009)
χ_M	Manufacturing variable innovation cost	0.060 (0.006)	0.055 (0.006)
ω_E	Coalfield innovation cost loading	0.0040 (0.0006)	0.0045 (0.0007)
λ_0	Baseline technological catch-up rate	0.160 (0.014)	0.175 (0.016)
δ	Spatial knowledge decay rate	1.350 (0.097)	1.245 (0.097)
Panel C. Infrastructure investment and expectations			
θ	Diagnostic expectations parameter	0.739 (0.090)	0 [fixed]
m_I	Infrastructure construction cost scale	0.023 (0.006)	0.020 (0.005)
χ_I	Infrastructure adjustment cost convexity	1.920 (0.233)	1.579 (0.213)
Panel D. Aggregate productivity process			
ρ_A	AR(1) persistence: Agriculture	0.168 (0.042)	0.174 (0.043)
ρ_M	AR(1) persistence: Manufacturing	0.533 (0.124)	0.561 (0.108)
ρ_T	AR(1) persistence: Transport	0.303 (0.076)	0.318 (0.079)
σ_A	Innovation s.d.: Agriculture	0.067 (0.004)	0.070 (0.004)
σ_M	Innovation s.d.: Manufacturing	0.039 (0.002)	0.042 (0.002)
σ_T	Innovation s.d.: Transport	0.023 (0.002)	0.025 (0.002)
ϱ_{AM}	Innovation correlation: (A, M)	−0.058 (0.015)	−0.082 (0.020)
ϱ_{AT}	Innovation correlation: (A, T)	0.094 (0.024)	0.077 (0.019)
$\varrho_{MT\|A}$	Partial innovation correlation: $(M, T \mid A)$	0.337 (0.084)	0.373 (0.093)

Notes: SMM estimates minimizing (77) across $M = 30$ moments (Table 6) under diagnostic (DE, $\theta \geq 0$, $p = 26$) and rational (RE, $\theta = 0$, $p = 25$) expectations across $B = 1{,}000$ common shock histories, with misspecification-robust sandwich standard errors (S.D.11) in parentheses. The wedge scale $\bar{\psi}_M = 0.251$ (SE = 0.032) is calibrated via (76), the eight 1710 productivity spline coefficients are concentrated out on the 1710 occupational cross-section, and fixed parameters are in Table 3.

Table 6: Empirical Moments and Model Fit

Empirical moment	Window	Data	DE	RE
Panel A. Aggregate growth and structural transformation				
Log real GDP trend (%/yr)	1710–1881	1.473 (0.210)	1.425	1.364
Log relative output $\ln(Y_M/Y_A)$ trend (%/yr)	1710–1881	1.085 (0.572)	0.970	0.799
Log relative price $\ln(p_M/p_A)$ trend (%/yr)	1710–1870	−0.757 (0.616)	−0.632	−0.496
Agricultural employment share trend (pp/yr)	1710–1881	−0.146 (0.023)	−0.139	−0.116
Tertiary employment share trend (pp/yr)	1710–1881	0.060 (0.007)	0.058	0.055
Panel B. Transport network growth and spatial dispersion				
Log transport stock: linear trend (%/yr)	1710–1881	1.948 (0.103)	1.943	1.758
Log transport stock: quadratic trend (×100)	1710–1881	2.923 (0.217)	2.952	2.144
Log transport stock: 1830–1850 growth (%/yr)	1830–1850	7.470 (0.759)	7.410	4.801
Log transport stock: post-1850 slope change (%/yr)	1830–1881	−5.101 (0.881)	−5.060	−2.461
Cross-band variance of log transport stock	1710–1881	0.469 (0.148)	0.437	0.410
Panel C. Cross-sectional employment covariances				
Manufacturing share–latitude covariance	1710–1881	0.077 (0.026)	0.070	0.065
Manufacturing share–energy covariance	1710–1881	−0.070 (0.024)	−0.064	−0.059
Spatial autocovariance of manufacturing share	1710–1881	0.030 (0.006)	0.028	0.026
Cross-band variance of tertiary share	1710–1881	0.0029 (0.0011)	0.0027	0.0025
Panel D. HP-filtered log sectoral output moments ($\times 10^4$)				
Variance: Agriculture (A)	1710–1881	34.79 (4.60)	35.81	39.80
Variance: Manufacturing (M)	1710–1881	9.64 (1.40)	9.96	12.12
Variance: Transport (T)	1710–1881	5.66 (0.93)	5.86	7.37
First-order autocovariance: Agriculture (A)	1710–1881	−7.27 (2.41)	−6.86	−6.75
First-order autocovariance: Manufacturing (M)	1710–1881	−0.49 (0.68)	−0.35	−0.08
First-order autocovariance: Transport (T)	1710–1881	−0.63 (0.39)	−0.55	−0.39
Cross-sector covariance: (A, M)	1710–1881	−2.59 (1.43)	−2.89	−3.89
Cross-sector covariance: (A, T)	1710–1881	−0.55 (0.83)	−0.71	−1.20
Cross-sector covariance: (M, T)	1710–1881	4.74 (0.78)	4.95	6.34
Panel E. Infrastructure investment dynamics				
Standard deviation of cyclical investment, $\sigma(c_t^I)$	1710–1880	0.431 (0.043)	0.427	0.345
Investment on current output cycle (b_0)	1711–1880	2.954 (1.431)	3.091	2.063
Investment on lagged output cycle (b_1)	1711–1880	1.585 (1.257)	1.662	3.142
First-order autocorrelation of investment (ρ_1)	1711–1880	−0.325 (0.074)	−0.344	−0.183
Lead output cycle on investment (b_{lead})	1711–1880	−0.0051 (0.0039)	−0.0049	−0.0035
County investment on population growth, lag 1 (g_1)	1791–1851	−0.162 (0.050)	−0.190	−0.082
County investment on population growth, lag 2 (g_2)	1791–1851	−0.496 (0.050)	−0.484	−0.144

Notes: Empirical moments (with source-bootstrap or Newey–West HAC standard errors in parentheses; Online Appendix S.D.8 and Table S.D.2) and simulated means across $B = 1{,}000$ common shock histories at the DE and RE estimates in Table 5. Sources: Broadberry et al. (2022) and Feinstein (1972) (Panels A, D), Shaw-Taylor et al. (2019) and Keibek (2017) (Panels A, C), and the measured transport stock (75) (Panels B, E).

D Aggregate Takeoff and Spatial Reallocation

Figures 7 and 8 plot the historical transition alongside the mean simulated trajectories and 5th–95th percentile bands across 1,000 Monte Carlo draws under DE and RE. Total real GDP rose 14.33-fold between 1710 and 1881, averaging 1.56 percent per year (Figure 7, Panel a), as annual growth accelerated from 1.02 percent in 1710–1760 and 1.16 percent in 1760–1800 to 1.60 percent in 1800–1830 and 2.37 percent in 1830–1881. Because no moment of the real wage series enters the SMM objective in Table 6, Panel (b) of Figure 7 provides an untargeted check on the model's price and wage predictions. Historically, real wages $w/\mathcal{P}$ rose modestly in the early eighteenth century, plateaued at 1.35–1.55 times their 1710 level during the late eighteenth and early nineteenth centuries—the "Engels' pause" of Allen (2009)—and then accelerated after 1820 to 3.28 times their 1710 level by 1881 as transport integration and industrial productivity gains lowered delivered prices. The industrial-to-agricultural output ratio Y_M/Y_A (Panel c) follows the same two-phase timing, drifting to 0.60 times its 1710 level in 1781 as population growth raised food demand before accelerating at 2.61 percent per year after 1813 to 5.77 times its 1710 level by 1881.[8] The estimated DE model reproduces this two-phase acceleration in aggregate output, Y_M/Y_A, and the untargeted real wage path $w/\mathcal{P}$ (1.41 in 1760, 1.75 in 1820, and 3.11 in 1881), whereas the restricted RE trajectory matches the eighteenth-century plateau (1.35 in 1760 and 1.47 in 1820) but stalls at 1.72 by 1881 alongside a muted rise in Y_M/Y_A.

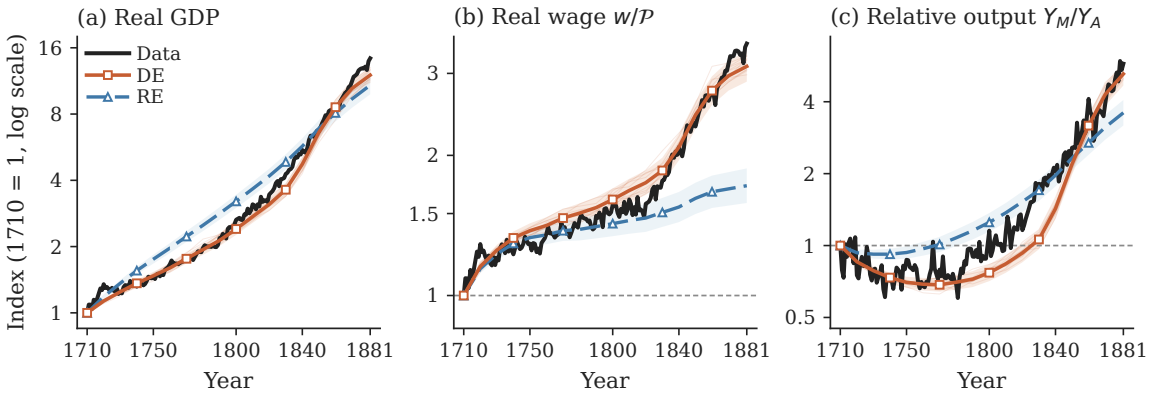


Figure 7: Aggregate Transition, 1710–1881

Notes: Historical series and simulated means (with 5th–95th percentile bands across 1,000 draws) under diagnostic (DE, $\widehat{\theta} = 0.739$) and restricted rational (RE, $\theta = 0$) expectations (1710 = 1, log scale). Panels (a) and (c) plot real GDP and relative industrial output Y_M/Y_A (Broadberry et al., 2022; Feinstein, 1972); Panel (b) plots the untargeted aggregate real wage $w/\mathcal{P}$ (Broadberry et al., 2022; Clark, 2002).

Transport capacity expanded nonlinearly over the sample. The measured stock (75) rose 27.70-fold between 1710 and 1881 (Figure 8, Panel c). Before 1830 it grew at 0.99 percent per year to 3.28 times its 1710 level, with turnpike-era bursts of 1.79 percent per year in 1750–1770 and 1.68 percent in the 1790s. Railway construction raised annual log growth to 5.90 percent in the 1830s and 8.43 percent in the 1840s (7.47 percent per year over 1830–1850), lifting the stock to 5.91 by 1840 and 13.73 by 1850, before annual growth slowed by 5.10 percentage points to 2.37 percent over 1850–1881 (24.36 in 1870). Disciplined by Panel B of Table 6, the DE path tracks both gradual eighteenth-century accumulation and the 1830–1850 surge and post-1850 moderation, whereas the RE path grows too smoothly and misses the mid-century acceleration. The 1881 band profile (Panel d) is

[8] The relative price p_M/p_A (Online Appendix Figure S.D.1, Panel c) mirrors this shift inversely, remaining above its 1710 level through much of the eighteenth century (peaking at 1.47 in 1733) before falling below unity after 1799 to 0.71 in 1800, 0.59 in 1830, 0.39 in 1855, and 0.49 in 1870.

multi-peaked: the West Riding and Lancashire band carried 53.8 times the 1710 national mean, followed by London (46.0) and the Midlands–Northern industrial bands (35.1–40.1), while Cornwall (3.5) and Northumberland (8.2) remained least served. Cross-band log-stock variance $V_{T,y}$ fell from 0.713 in 1710 to 0.295 in 1881 in the data (0.658 to 0.240 under DE; Figure S.D.1, Panel f), with a sample mean of $\overline{V}_T = 0.469$ (0.437 under DE and 0.410 under RE; Panel B). Under DE, counties build more aggressively during the canal and railway booms across the Midlands and Northern corridor (bands 12–22) than under RE, matching the 1881 cross-sectional peaks.

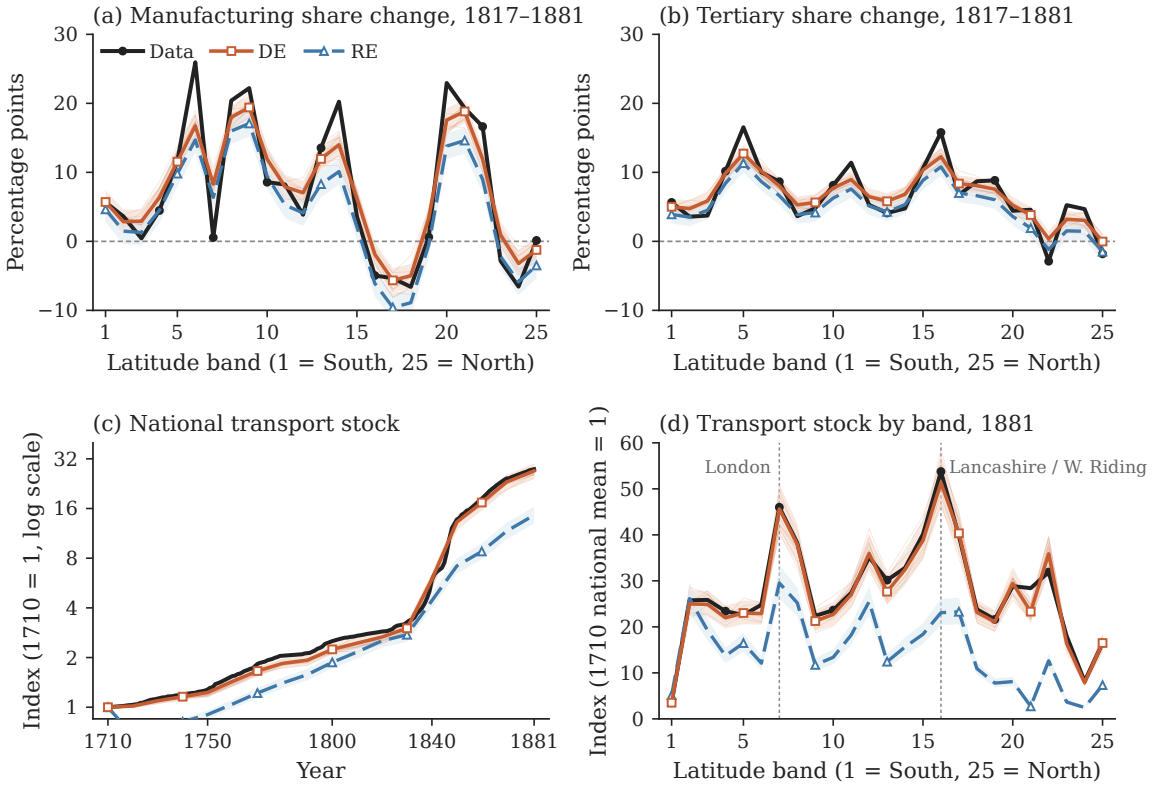


Figure 8: Spatial Reallocation and Infrastructure, 1710–1881

Notes: Historical series and simulated means (with 5th–95th percentile bands across 1,000 draws) under DE ($\widehat{\theta} = 0.739$) and RE ($\theta = 0$). Panels (a)–(b) plot 1817–1881 percentage-point changes in manufacturing and tertiary employment shares across the 25 latitude bands (1 = South, 25 = North; Online Appendix S.B); Panels (c)–(d) plot the national mean transport stock (75) (1710 = 1, log scale) and the 1881 band profile (1710 national mean = 1).

Panels (a)–(b) of Figure 8 compare the 1817–1881 change in manufacturing and tertiary employment shares across the twenty-five latitude bands with the DE and RE predictions (Online Appendix S.B). As the national manufacturing share rose from 49.9 to 62.1 percent, industrial employment diffused across space: the manufacturing share rose in 20 of 25 bands and its change correlates negatively with the 1817 share (−0.81), with 20–26 percentage point gains in initially peripheral bands (bands 6, 8, and 20–22) and 5–7 point declines in the Lancashire–Yorkshire core (bands 16–18, which remained above 79 percent manufacturing in 1881). Tertiary employment expanded in 23 of 25 bands as the national transport and distribution share nearly doubled from 9.5 to 18.5 percent, led by the Southern, London, and Lancashire–West Riding corridors. Although the estimation targets only aggregate occupational series and dated spatial covariances in Panels A and C, the DE model captures both manufacturing diffusion and tertiary expansion more closely

than RE.[9]

E Counterfactual Analysis: Diagnostic versus Rational Expectations

Re-estimating the model under $\theta = 0$ alters technology and construction-cost parameters alongside beliefs. To isolate diagnostic expectations, the belief-neutralization counterfactual (R | DE) holds all other parameters at their diagnostic estimates, together with the 1710 state, statutory project opportunities, and primitive shock draws $\{u_b\}_{b=1}^{B}$, sets $\theta = 0$, and recomputes county investment policies $\mathbf{x}_t^0$ and the general equilibrium path from 1710 to 1881.

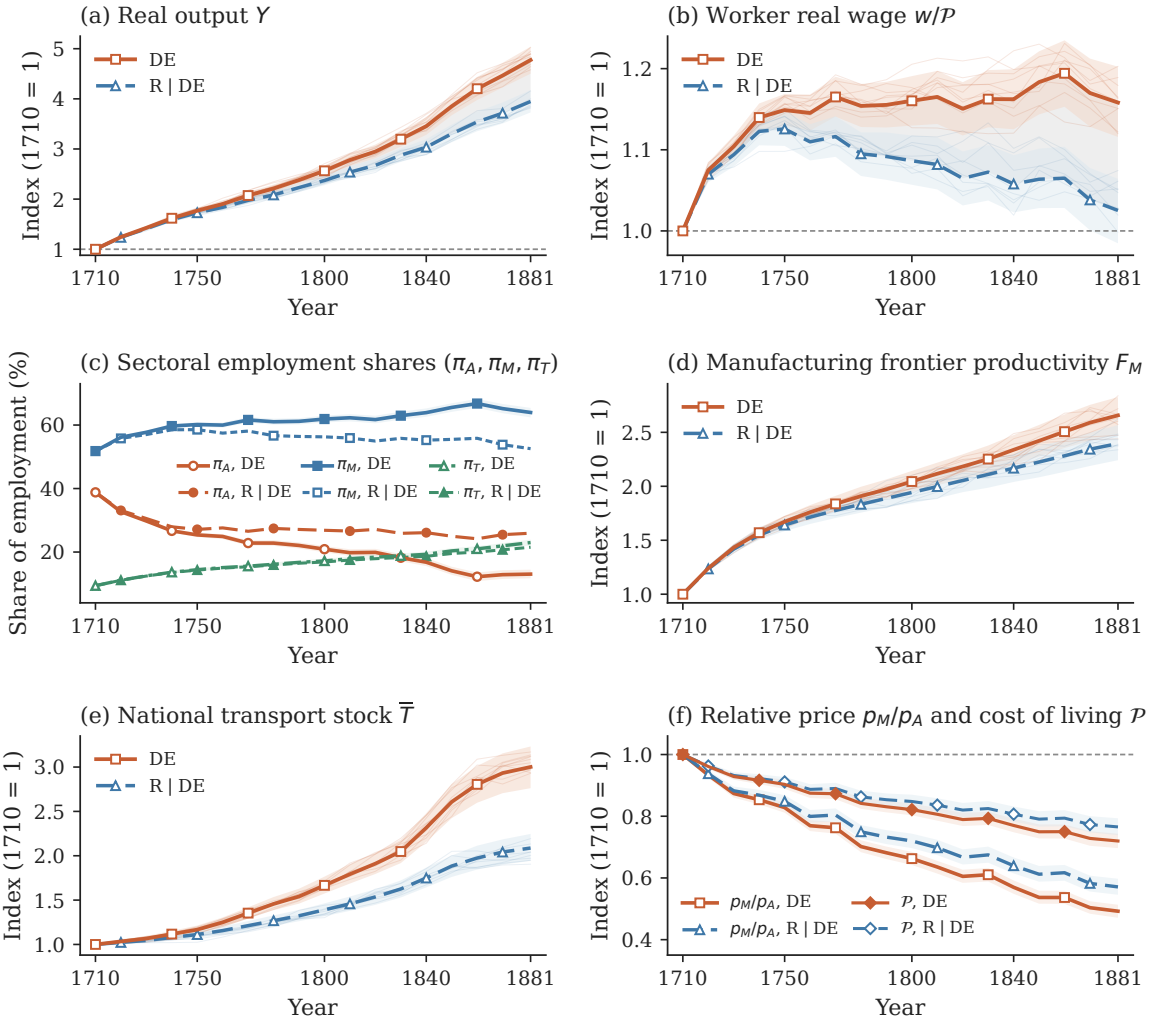


Figure 9: Counterfactual Effects of Diagnostic Expectations, 1710–1881

Notes: Simulated means and 5th–95th percentile bands (1,000 draws) under the baseline diagnostic economy (DE, $\widehat{\theta} = 0.739$) and the belief-neutralization counterfactual (R | DE: $\theta = 0$ holding all other parameters and shock histories fixed). Panels (a), (b), (d), (e), and (f) report balanced-growth-detrended indices (1710 = 1) for real output Y, real wage $w/\mathcal{P}$, manufacturing frontier F_M, transport stock $\overline{T}$, and prices ($p_M/p_A, \mathcal{P}$); Panel (c) reports sectoral employment shares π_A, π_M, π_T (percent).

By Proposition 2, household and firm decisions and prices at each date are invariant to θ given county investment, so every difference between the DE and R | DE paths originates in county

[9]Online Appendix S.D (Figure S.D.1) plots the complete historical and simulated series for every moment block.

construction and propagates through the installed network. Construction absorbs manufactures when built; completed capacity lowers route losses $\kappa(T)$ and expands carrier service capacity at later dates; and, by Proposition 3, higher capacity accelerates spatial knowledge catch-up.

Figure 9 and Table 7 report the counterfactual.[10] By 1881, diagnostic expectations expand the aggregate transport stock $\overline{T}$ by +43.79 percent (3.001 versus 2.087; Panel e), with regional gains of +29.53 percent in the Southern belt (Counties 1–5), +54.47 percent in the Midlands–North corridor (Counties 6–9), and +61.33 percent in the Northern border (County 10). This network expansion raises 1881 real output Y (and output per worker y) by +20.96 percent (4.774 versus 3.947; Panel a) and lifts the population-weighted real wage $w/\mathcal{P}$ by +12.97 percent nationally (1.158 versus 1.025; +12.78 to +13.59 percent across regions; Panel b). Larger route capacity accelerates structural transformation (Trew, 2020), lowering the 1881 agricultural employment share by −12.9 percentage points (13.0 versus 25.9 percent) and shifting +11.4 percentage points into manufacturing (64.0 versus 52.6 percent) and +1.5 percentage points into tertiary services (23.0 versus 21.5 percent; Panel c). Faster spatial diffusion lifts manufacturing frontier productivity F_M by +10.90 percent (2.657 versus 2.396; Panel d), while cheaper distribution and higher manufacturing productivity lower p_M/p_A by −13.79 percent (0.492 versus 0.571) and the Stone–Geary price index $\mathcal{P}$ by −5.97 percent (0.720 versus 0.765; Panel f).

Two features of Table 7 highlight the model's spatial mechanisms. First, in Panel B, the counterfactual percentage gain in the available frontier $F_{Mt}(\ell) = \sup_{r\in[0,1]} e^{-\delta|\ell-r|}A^{+}_{Mt}(r)$ is spatially uniform at +10.90 percent across all three regions because the spatial supremum scales proportionally with the northern industrial peak. Local applied manufacturing productivity $A_{Mt}(\ell)$, governed by the transport-dependent catch-up rate $\lambda_t(\ell) = \lambda_0[T_t(\ell)/T_{t_0}(\ell)]^{1-\alpha_T}$ in (39)–(40), expands most where route buildout is largest (+7.84 percent in the Southern belt, +13.42 percent in the Midlands–North corridor, and +14.18 percent in the Northern border, or +11.65 percent nationally). Second, decomposing the +20.96 percent 1881 output gain sequentially across the three uninternalized social return channels of Section III.F shows why static "social savings" calculations (Fogel, 1964) capture only a fraction of the long-run return to connective infrastructure: direct carrier and iceberg transit savings holding spatial sorting and technology at their R | DE paths (Donaldson and Hornbeck, 2016; Fogel, 1964) account for +5.42 percentage points (25.9 percent of the gain); spatial labor reallocation into wedge-distorted manufacturing ($\psi_M(\ell) > 0$; Baqaee and Farhi 2020; Hornbeck and Rotemberg 2024; Hsieh and Klenow 2009) accounts for +6.18 percentage points (29.5 percent); and transport-mediated spatial knowledge diffusion and induced innovation (65) (Desmet and Rossi-Hansberg, 2014; Mokyr, 2005; Nelson and Phelps, 1966; Romer, 1990) account for +9.36 percentage points (44.7 percent).

[10] Level indices in Figure 9 and Table 7 are in balanced-growth-detrended units (1710 = 1), and employment shares are model-domain shares $\mathcal{L}_o(t)/\bar{L}_t$ (S.D.2). Because exogenous package-scale and frontier drift factors match across DE and R | DE, percentage differences $100\,(\text{DE/R} \mid \text{DE} - 1)$ are identical in un-detrended units.

Table 7: Counterfactual Effects of Diagnostic Expectations in 1881

1881 outcome (1710 = 1, except shares)	DE ($\widehat{\theta} = 0.739$)	R | DE ($\theta = 0$)	Difference
Panel A. Aggregate outcomes			
Real output (Y)	4.774	3.947	+20.96%
Population-weighted real wage ($w/\mathcal{P}$)	1.158	1.025	+12.97%
Agricultural employment share (π_A, %)	13.0	25.9	−12.9 pp
Manufacturing / tertiary shares (π_M/π_T, %)	64.0 / 23.0	52.6 / 21.5	+11.4 / +1.5 pp
Manufacturing frontier productivity (F_M)	2.657	2.396	+10.90%
Aggregate transport stock ($\overline{T}$)	3.001	2.087	+43.79%
Relative price of manufactures (p_M/p_A)	0.492	0.571	−13.79%
Stone–Geary price index ($\mathcal{P}$)	0.720	0.765	−5.97%
Panel B. Regional outcomes ($w/\mathcal{P}$ / F_M / $\overline{T}$)			
Southern belt (Counties 1–5)	1.188 / 1.888 / 2.470	1.053 / 1.703 / 1.907	+12.78% / +10.90% / +29.53%
Midlands–North corridor (Counties 6–9)	1.156 / 3.299 / 3.549	1.023 / 2.975 / 2.298	+13.00% / +10.90% / +54.47%
Northern border (County 10)	1.116 / 3.935 / 3.463	0.982 / 3.548 / 2.146	+13.59% / +10.90% / +61.33%

Notes: 1881 outcomes averaged across $B = 1{,}000$ common shock histories under baseline diagnostic expectations (DE, $\widehat{\theta} = 0.739$) and the belief-neutralization counterfactual (R | DE, $\theta = 0$ holding all other parameters and shock histories fixed). Level indices are in balanced-growth-detrended units (1710 = 1; percentage differences $100\,(\text{DE}/\text{R} \mid \text{DE} - 1)$ are identical in historical units), and employment shares are in percent (differences in percentage points).

V Conclusion

Economic historians and financial economists have long viewed the recurrent investment manias of the British Industrial Revolution—and subsequent general-purpose network buildouts from electrification to fiber-optic telecommunications (Bresnahan and Trajtenberg, 1995)—through two seemingly incompatible lenses. Financial histories emphasize promoter over-extrapolation, post-boom construction reversals, and disappointing realized yields (Arnold and McCartney, 2011; Campbell, 2012; Keynes, 1936; Kindleberger, 1978; Odlyzko, 2010; Shaen, 1847; Shiller, 2000); growth and spatial histories emphasize that the physical networks left behind permanently integrated national markets and accelerated industrialization (Bogart, 2014; Donaldson and Hornbeck, 2016; Fogel, 1964; Mokyr, 2005; Trew, 2020). Embedding a Markov perfect county infrastructure game under diagnostic expectations into a continuous-space endogenous growth model of England and Wales over 1710–1881 shows that both regularities are two sides of a single second-best mechanism: *corrective overoptimism* (Lipsey and Lancaster, 1956). Because decentralized promoters appropriate only local land rents inside their jurisdictions, rational expectations chronically underinvest relative to the social returns from inter-county transit savings, labor reallocation into wedge-distorted manufacturing, and spatial technology diffusion. Diagnostic overreaction to favorable productivity news induces promoters to build aggressively during booms, and physical irreversibility (Arrow, 1968; Dixit and Pindyck, 1994) locks in the resulting route capacity long after expectations revert.

Estimating the model via the Simulated Method of Moments yields a diagnosticity parameter of $\widehat{\theta} = 0.739$ (SE = 0.090), matching the $\theta \approx 0.74$–0.91 range documented in financial and macroeconomic expectations (Bordalo et al., 2018, 2026, 2019) and identified by the coexistence of strong boom-time investment comovement with sharp post-boom reversals. In the belief-neutralization counterfactual (R | DE), diagnostic expectations expand the 1881 aggregate transport stock by +43.79 percent (+54.47 percent in the Midlands–North corridor), accelerate structural transformation out of agriculture by −12.9 percentage points, lift manufacturing frontier productivity by +10.90 percent, and raise 1881 real output by +20.96 percent and worker real wages by +12.97 percent.

These findings carry three broader implications. First, they recast the "social savings" debate initiated by Fogel (1964): static freight-cost savings capture only about one-quarter (25.9 percent, or +5.42 percentage points) of the long-run output gain from belief-driven network expansion, while nearly three-quarters arises from reallocating labor into wedge-distorted manufacturing (29.5 percent, or +6.18 percentage points; Baqaee and Farhi 2020; Hornbeck and Rotemberg 2024; Hsieh and Klenow 2009; Restuccia and Rogerson 2008) and accelerating the spatial diffusion of frontier discoveries (44.7 percent, or +9.36 percentage points; Aghion and Howitt 1992; Lucas 1988; Krugman 1991; Mokyr 2005; Nelson and Phelps 1966; Romer 1990). Second, they establish a primitive boundary condition separating *corrective* from *destructive* belief overreaction (Bordalo et al., 2018, 2026, 2020; Kahneman and Tversky, 1972): when extrapolative beliefs fund non-durable consumption or zero-sum speculation without technological externalities, the bust leaves balance-sheet distress with little productive legacy; when overoptimism funds irreversible connective capital whose social returns exceed private appropriation, private financial disappointment coexists with permanent productivity gains. Finally, corrective overoptimism highlights a comparative institutional trade-off (North, 1990): continental European states addressed infrastructure underprovision through centralized planning and sovereign dividend guarantees (Bogart, 2014; Milward and Saul, 1973), whereas Britain and the United States relied on decentralized private promotion amplified by speculative manias (Janeway, 2018; Morck, 2022). Policies that lean uniformly against investor exuberance—or insulate incumbent network providers from entry—risk suppressing the decentralized investment waves that overcome chronic underprovision.

subsection

A Proofs of Main Results

A.1 Proof of Proposition 1

Proof. Because portfolio shares are residence independent, annual relocation is costless, and households are atomistic, continuation utility $\mathbb{E}_t^\theta[W_{t+1}]$ does not depend on current residence ℓ_t. The worker's Bellman equation therefore separates into $W_t = \max_{\ell_t \in [0,1]} \mathbb{E}_t^\theta[u_t(\ell_t)] + \beta \mathbb{E}_t^\theta[W_{t+1}]$, so occupied locations maximize $\mathbb{E}_t^\theta[u_t(\ell)]$ at the common value $\bar{u}_t$ and unoccupied locations satisfy $\mathbb{E}_t^\theta[u_t(\ell)] \le \bar{u}_t$. [← Back to Proposition 1] □

A.2 Proof of Proposition 2

Proof. In (10), the likelihood ratio $\Lambda_t(\mathbf{z}_{t+1})^\theta$ depends only on future aggregate shocks $\mathbf{z}_{t+1}$, which are independent of current private discoveries $G_{it}(\ell, \omega)$ conditional on $(\Omega_t, \mathbf{x}_t)$. Consequently, $\Lambda_t(\mathbf{z}_{t+1})^\theta$ factors out of the numerator and denominator of $\mathbb{E}_t^\theta[X_t]$ for every date-t variable $X_t(\omega)$, reducing all within-period expectations to $\mathbb{E}_t^\theta[X_t] = \mathbb{E}_t[X_t]$ and leaving the conditional dated equilibrium and physical state transitions invariant to θ once $\mathbf{x}_t$ is fixed. [← Back to Proposition 2] □

A.3 Proof of Proposition 3

Proof. Let $(\mathcal{D}A^+)(\ell) = \sup_{r \in [0,1]} e^{-\delta|\ell - r|} A^+(r)$ and $F = \mathcal{D}A^+$. Taking $r = \ell$ gives $A^+ \le F$, and the triangle inequality $|\ell - s| \le |\ell - r| + |r - s|$ implies $\mathcal{D}F = F$. Because $A' = e^{-\lambda}A^+ + (1 - e^{-\lambda})F$ is a convex combination of A^+ and F for $\lambda \ge 0$, we have $A^+ \le A' \le F$ and hence $F = \mathcal{D}A^+ \le \mathcal{D}A' \le \mathcal{D}F = F$. Without further discoveries, F is unchanged and iterating $F - A' = e^{-\lambda}(F - A^+)$ gives (41); since $\lambda_t(\ell)$ increases in $T_t(\ell)$ and $F - A^+ \ge 0$, higher $T_t(\ell)$ weakly raises $A_{i,t+1}(\ell)$. [← Back to Proposition 3] □

A.4 Proof of Proposition 4

Proof. Since $\mathcal{X}_t = \prod_j [0, \bar{x}_{jt}]$ is nonempty, compact, and convex and each Π_{jt}^θ is continuous in $\mathbf{x}$ and concave in x_{jt}, Kakutani's fixed-point theorem yields a nonempty compact Nash equilibrium set $\mathcal{N}_t^\theta$. When Π_{jt}^θ is differentiable, concavity makes the variational inequality

$$g_{jt}^\theta(\mathbf{x}^*)(u - x_j^*) \le 0 \quad \forall\, u \in [0, \bar{x}_{jt}], \quad j = 1, \dots, J, \qquad g_{jt}^\theta = \partial_j \Pi_{jt}^\theta, \tag{A.1}$$

necessary and sufficient. Under (62), each best response $b_j(\mathbf{y}_{-j})$ is unique, and comparing $u = b_j(\mathbf{y}_{-j})$ with $v = b_j(\mathbf{z}_{-j})$ in (A.1) gives $(u - v)\{g_j(u, \mathbf{y}_{-j}) - g_j(v, \mathbf{z}_{-j})\} \ge 0$, hence $m_j|u - v| \le \sum_{k \ne j} L_{jk}|y_k - z_k|$. In the weighted maximum norm $\|\mathbf{q}\|_v = \max_j |q_j|/v_j$, we obtain $\|\mathbf{b}(\mathbf{y}) - \mathbf{b}(\mathbf{z})\|_v \le \gamma_I \|\mathbf{y} - \mathbf{z}\|_v$ with $\gamma_I < 1$ by (63), so Banach's fixed-point theorem yields a unique Nash equilibrium and geometric convergence of best responses. [← Back to Proposition 4] □

A.5 Proof of Proposition 5

Proof. Fix a realization ω. Writing local carrier demand as $D = \sum_i r_i(Y_i + |H_i|)$ and using local labor clearing $n = \sum_i e_i L_i + L_T$, summing the local ledgers (69)–(71) gives

$$R^{\text{net}} + \Pi_F + \Pi_T + \Pi_\psi = \sum_i p_i Y_i + \tau_{j(\ell)} \sum_i r_i |H_i| + \tau_{j(\ell)}(Y_T - D) - wn - \sum_i e_i \Psi_i - p_M I. \tag{A.2}$$

Because τ_j is constant within each county $\mathcal{C}_j$, county service clearing (50) implies

$$\int_{\mathcal{C}_j} \tau_j Y_T \, d\ell = \tau_j K_j = \tau_j D_j = \int_{\mathcal{C}_j} \tau_j D \, d\ell, \tag{A.3}$$

so integrating (A.2) over $[0,1]$ yields

$$b_t \bar{L}_t = \int_0^1 (R^{\text{net}} + \Pi_F + \Pi_T + \Pi_\psi) d\ell = \int_0^1 \Big[\sum_i p_i Y_i + \tau_{j(\ell)} \sum_i r_i |H_i| - wn - \sum_i e_i \Psi_i - p_M I \Big] d\ell. \tag{A.4}$$

Meanwhile, combining goods conservation (68) with $H_i \partial_\ell p_i = (\kappa p_i + r_i \tau_{j(\ell)}) |H_i|$ from spatial no-arbitrage (44) cancels the physical loss terms $\pm \kappa p_i |H_i|$ in $\partial_\ell (p_i H_i) = p_i \partial_\ell H_i + H_i \partial_\ell p_i$, giving $\partial_\ell (p_i H_i) = p_i (Y_i - n c_i - e_i \Psi_i / p_i - I_i) + r_i \tau_{j(\ell)} |H_i|$ ($I_A = 0$, $I_M = I$). Integrating over $[0,1]$ under $H_i(0) = H_i(1) = 0$, summing over $i \in \{A, M\}$, and substituting (A.4) gives $\int_0^1 \sum_i p_i n c_i \, d\ell = \int_0^1 wn \, d\ell + b_t \bar{L}_t$. [← Back to Proposition 5] □

A.6 Proof of Proposition 6

A.6.1 Part 1: Firm attainment, active-research threshold, and pure-site sorting

Fix $\ell \in [0,1]$ and sector $i \in \{A, M\}$ at bounded quotes $0 \le p_i(\omega) \le \bar{p} < \infty$, $\tau_{j(\ell)}(\omega) \ge 0$, and $w > 0$, and write $m_i(\phi) = \mathbb{E}_t[(p_i - r_i \tau_{j(\ell)})_+ G_i(\phi)]$.

LEMMA A.1: *Under (21)–(22) with $a_i > 1/(1-\alpha_i) > 1$, $m_i(\phi)$ is continuous on $[0,1]$ and satisfies*

$$\bar{m}_i := \sup_{0 \le \phi \le 1} m_i(\phi) \le \bar{p} \sup_{0 \le \phi \le 1} \mathbb{E}[G_i(\phi)] = \bar{p} \, \frac{a_i}{a_i - 1} < \infty. \tag{A.5}$$

For each $\phi \in [0,1)$, optimal production labor is unique and given by

$$L_i^*(\phi) = \left[\frac{\alpha_i Z_i m_i(\phi)}{(1+\psi_i) w} \right]^{1/(1-\alpha_i)}, \tag{A.6}$$

and the profiled bid rent

$$R_i(\phi; w) = \Lambda_i w^{-\frac{\alpha_i}{1-\alpha_i}} m_i(\phi)^{1/(1-\alpha_i)} - \Psi_i(\phi; w), \qquad \Lambda_i = (1-\alpha_i) Z_i \left[\frac{\alpha_i Z_i}{1+\psi_i} \right]^{\alpha_i/(1-\alpha_i)}, \tag{A.7}$$

is finite and attained on $[0,1)$.

Proof. Since $G_i(\phi; \ell, \omega) = 1 + \mathbf{1}_{\{U_i < \phi\}}(V_i - 1)$ with U_i independent of (V_i, p_i, τ_j), $m_i(\phi) = m_i(0) + \phi \, \mathbb{E}_t[(p_i - r_i \tau_{j(\ell)})_+ (V_i - 1)] = m_i(0)[1 + \kappa_i(\ell) \phi]$ is affine on $[0,1]$ with $\mathbb{E}[G_i(\phi)] = 1 + \phi/(a_i - 1) \le a_i/(a_i - 1) < \infty$, giving (A.5). Strict concavity in L^{α_i} yields (A.6)–(A.7), and $\Psi_i(\phi; w) \ge w \chi_i / (1 - \phi) \to \infty$ as $\phi \uparrow 1$ ensures attainment on $[0,1)$. □

At an interior optimum $\tilde{\phi}_i \in (0,1)$, the envelope theorem gives

$$Z_i L_i^*(\tilde{\phi}_i)^{\alpha_i} \, \partial_\phi \mathbb{E}_t \Big[(p_i - r_i \tau_{j(\ell)})_+ G_i(\tilde{\phi}_i) \Big] = \partial_\phi \Psi_i(\tilde{\phi}_i; w) = \frac{w \chi_i}{(1 - \tilde{\phi}_i)^2}, \tag{A.8}$$

matching (31). For $w_2 > w_1$ with optimal plans (ϕ_2, L_2) and (ϕ_1, L_1), writing $R_i(\phi; w, \ell)/w = \pi_i^0(w, \ell)[1 + \Gamma_i(\phi; \ell)] - C_i(\phi)$ with $C_i(\phi) = \Psi_i(\phi; w, \ell)/w$ and summing the revealed-preference inequalities gives

$$[\pi_i^0(w_2, \ell) - \pi_i^0(w_1, \ell)] \, [\Gamma_i(\phi_2; \ell) - \Gamma_i(\phi_1; \ell)] \ge 0. \tag{A.9}$$

Since $\pi_i^0(w_2,\ell) < \pi_i^0(w_1,\ell)$ and $\Gamma_i(\cdot;\ell)$ is strictly increasing, $\phi_2 \le \phi_1$, $m_i(\phi_2;\ell) \le m_i(\phi_1;\ell)$, and $L_2 = L_i^*(\phi_2;w_2,\ell) < L_i^*(\phi_1;w_1,\ell) = L_1$: thus $\phi_i^*(w,\ell)$ is weakly decreasing and $L_i^*(\phi_i^*(w,\ell);w,\ell)$ is strictly decreasing in w. For $\chi_i > \bar{\chi}_{it} \equiv \frac{\alpha_i}{2(1-\alpha_i)^2}\sup_{\ell,w,\phi\in(0,1)}(1-\phi)^3\kappa_i(\ell)^2\Lambda_i(\ell)w^{-1/(1-\alpha_i)}$ $m_i(\phi;\ell)^{(2\alpha_i-1)/(1-\alpha_i)}$, $\partial^2_{\phi\phi}R_i(\phi;w,\ell) < 0$ on $(0,1)$, so at most one interior stationary point $\tilde{\phi}_i(w,\ell)$ exists, and (32) uniquely selects $\phi_i^*(w,\ell) \in \{0,\tilde{\phi}_i(w,\ell)\}$. Under Condition 2, $R_A(\ell) - R_M(\ell)$ crosses zero transversally at a unique $b_{ct} \in (0,1)$, assigning $[0,b_{ct}]$ to agriculture and $(b_{ct},1]$ to manufacturing. With e_{it} and ϕ_{it}^* fixed at every ℓ, (37) determines $A_{it}^+(\ell,\omega)$ pointwise, and Lemma S.C.1 (Online Appendix) ensures $F_{it}(\ell,\omega)$ in (38) is finite a.s. with finite mean everywhere on $[0,1]$.

A.6.2 Part 2: Existence and norm-contraction global uniqueness

Proof. On $\mathcal{W}_t \times \mathcal{P}_t \times \mathcal{M}_t$ (Condition 1), $G_i(\phi_i^*;\ell,\omega) \ge 1$ implies $L_i^*(\phi_i^*;w,\ell) \ge L_i^*(0;w,\ell) \propto w^{-1/(1-\alpha_i)}$ ($L_T^* \propto w^{-1/(1-\alpha_T)}$), whereas $\Psi_i(\phi_i^*;w,\ell) = O(w)$ vanishes as $w \downarrow 0$; hence aggregate labor demand exceeds $\bar{L}_t$ at $\underline{w}_t > 0$ and falls below $\bar{L}_t$ at $\overline{W}_t < \infty$. Given predetermined capacities and Condition 1's subsistence, construction, and service slack, maximizing the strictly concave Negishi-weighted Stone–Geary log-surplus $\int_0^1 \vartheta(\ell,\omega)n(\ell)[\eta\log(c_A(\ell,\omega)-\gamma)+(1-\eta)\log c_M(\ell,\omega)]\,d\ell$ at weights $\vartheta(\ell,\omega) = w(\ell)+b_t(\omega)-\gamma p_A(\ell,\omega)$ subject to (68) and $D_j(\omega) \le K_j$ (50) yields unique KKT spot prices $\widetilde{p}_i(\cdot,\omega) \in W^{1,\infty}([0,1])$ ($\widetilde{p}_A(1,\omega) = 1$) and county tolls $\widetilde{\tau}_j(\omega) \ge 0$. Since $|\partial_\ell \widetilde{p}_i| \le \kappa(T_t)\widetilde{p}_i + r_i\widetilde{\tau}_{j(\ell)}$ a.e., substituting $(\widetilde{p}(w),\widetilde{\tau}(w))$ into $\mathcal{T}_t^w$ in (73) defines a reduced wage operator $\widetilde{\mathcal{T}}_t^w : \mathcal{W}_t \to \mathcal{W}_t$ on $C([0,1])$ with equicontinuous image. By Arzelà–Ascoli and Schauder, $\mathcal{T}_t$ admits a fixed point $q^* = (w^*,p^*,\tau^*)$, which is unique under Condition 3 since $\|q_1-q_0\| = \|\mathcal{T}_t(q_1)-\mathcal{T}_t(q_0)\| \le \bar{\rho}_t\|q_1-q_0\|$ with $\bar{\rho}_t < 1$. □

A.6.3 Part 3: Constructive verification from primitives

Proof. Step 1 (Frictionless benchmark). When $\kappa_0 = r_A = r_M = m_I = 0$ and $f_i + \chi_i > \bar{f}_i \equiv \sup_{\ell,\phi\in(0,1)}(1-\phi)\pi_i^0(w_t^0,\ell)\Gamma_i(\phi;\ell)$, prices and wages equalize across space ($p_{At} \equiv 1$, $p_{Mt} \equiv p_{Mt}^0 > 0$, $\tau_{jt} \equiv 0$, $\phi_{it}^* \equiv 0$, $w_t \equiv w_t^0$). Opposite-signed log-knowledge slopes ($\partial_\ell \log A_{At} < 0 < \partial_\ell \log A_{Mt}$) make $\log(R_{At}(\ell)/R_{Mt}(\ell))$ strictly decreasing in ℓ, pinning down a unique cutoff $b_{ct}^0 \in (0,1)$ with $\partial_\ell[R_{At}(b_{ct}^0) - R_{Mt}(b_{ct}^0)] < 0$, while labor clearing and Stone–Geary expenditure shares determine $(w_t^0, p_{Mt}^0, b_{ct}^0)$ in closed form.

Step 2 (Uniform price and wage bounds). For any accumulated stock $T_t(\ell) = T_0(\ell) + s_{j(\ell),t}(\ell) \ge T_0(\ell)$, higher $s_{j(\ell),t}$ expands carrier capacity K_j and reduces route loss $\kappa_0 e^{-T_t}$, so county tolls are bounded by their initial-stock ceilings $\overline{\tau}_j(0)$ and bounded below in expectation by $\underline{\tau}_{jt} > 0$. Integrating (44) from $p_{At}(1,\omega) = 1$ across counties $j = J,\dots,1$ bounds $p_{it}(\ell,\omega) \in [\underline{p}_i,\overline{p}_i]$ and $w(\ell) \in [\underline{w}_{\text{nat}},\overline{W}_{\text{nat}}]$ uniformly across all stock paths $\{s_{jt}\}_{j=1}^J$, verifying Conditions 1 and 2.

Step 3 (Contraction under small frictions). Since b_{ct} and any active-research thresholds cross zero transversally, $\mathcal{T}_t$ is C^1 on $\mathcal{W}_t \times \mathcal{P}_t \times \mathcal{M}_t$ and contracts at the frictionless benchmark with modulus $\rho_0 < 1$ governed by diminishing returns to labor ($-1/(1-\alpha_i) < 0$) and Stone–Geary demand monotonicity. By continuity of $\mathcal{D}_t$, $\sup_q \|\mathcal{D}_t(q)\| \le \frac{1}{2}(1+\rho_0) < 1$ persists for small $(\kappa_0 e^{-T_0}, r_A, r_M, m_I)$ uniformly along every admissible infrastructure path, verifying Condition 3.
[← Back to Proposition 6] □

A.7 Proof of Proposition 7

A.7.1 Part 1: Recursive county contraction and state transitions

Let $K_{\mathbf{x}}^\theta$ be the one-period forecasting kernel on Ω under (10)–(12), with stage value $Q_j(\Omega,\mathbf{x};V) = R_j(\mathbf{x};\Omega) + \beta(K_{\mathbf{x}}^\theta V_j)(\Omega)$. The weight $W(\Omega_t) = 1 + \sum_{i\in\{A,M\}}\|A_{it}\|_\infty + \sum_{j=1}^J \overline{T}_{jt}$ has finite one-step

conditional mean $(K^{\theta}_{\mathbf{x}}W)(\Omega_t) \le \kappa W(\Omega_t)$ because $\sup_{\ell\in[0,1]} A_{i,t+1}(\ell) \le \sup_{\ell\in[0,1]} F_{it}(\ell)$ has finite conditional mean by Lemma S.C.1 (Online Appendix).

THEOREM A.1: *On a forecast-invariant state domain where each $\mathbf{x} \in \mathcal{X}_t$ has a unique measurable private equilibrium, let $W \ge 1$ be a measurable weight and $\mathcal{C}$ a closed class of county-value vectors equipped with the weighted C^1 route-stock norm $\|V\|_{W,1} = \max_j \sup_\Omega W(\Omega)^{-1}(|V_j(\Omega)| + \sum_{k=1}^{J} \overline{T}_k |\partial_{\overline{T}_k} V_j(\Omega)|)$. Suppose $|R_j(\mathbf{x};\Omega)| \le C_R W(\Omega)$ and $(K^{\theta}_{\mathbf{x}}W)(\Omega) \le \kappa W(\Omega)$. If for every $U, V \in \mathcal{C}$ the stage game has a unique measurable Nash profile $a_V(\Omega)$ satisfying $\|a_V(\Omega) - a_U(\Omega)\|_v \le L_a\|V - U\|_{W,1}$, the maximized payoff and its stock derivatives satisfy the Lipschitz bound with modulus $BW(\Omega)$ in $\|\cdot\|_v$, and the Nash Bellman operator $(\mathcal{B}^{\theta}V)_j(\Omega) = \max_{x_j} Q_j(\Omega, x_j, a_{V,-j}(\Omega); V)$ maps $\mathcal{C}$ into itself with $\beta\kappa + BL_a < 1$, then $\mathcal{B}^{\theta}$ has a unique fixed point $V^{\theta} \in \mathcal{C}$ and a unique Markov perfect policy $a_{V^{\theta}}$.*

Proof. Holding actions fixed, $Q_j(\Omega, \mathbf{x}; V) - Q_j(\Omega, \mathbf{x}; U)$ and its stock derivatives are bounded by $\beta\kappa W(\Omega)\|V - U\|_{W,1}$, while shifting rival actions from $a_{V,-j}$ to $a_{U,-j}$ (and applying the envelope theorem for $a_{V,j}$) adds at most $BW(\Omega)L_a\|V - U\|_{W,1}$. Thus $\|\mathcal{B}^{\theta}V - \mathcal{B}^{\theta}U\|_{W,1} \le (\beta\kappa + BL_a)\|V - U\|_{W,1}$, yielding a unique fixed point $V^{\theta} \in \mathcal{C}$. □

COROLLARY A.1: *Under Theorem A.1, every realization of the physical shock and discovery laws from Ω_0 generates a unique path of allocations, knowledge frontiers, productivity gaps, and route stocks.*

Proof. Composing the unique conditional dated equilibrium map with the unique county policy $a_{V^{\theta}}$ fixes the allocation and investment profile at each state, and iterating (9), (37)–(39), and (53) determines the realized path by induction. □

A.7.2 Part 2: Constructive verification on the rescaled stock corridor

Proof. Fix the $J = 10$ counties $\mathcal{C}_j = [(j-1)/10, j/10]$ and the primitive regime of Proposition 6. At date t with $n \ge 1$ periods remaining in horizon H, county j maximizes $\Pi^{(n),\theta}_{jt}(\mathbf{x}_t; \Omega_t) = R_{jt}(\mathbf{x}_t; \Omega_t) + \beta\mathbb{E}^{\theta}_t[V^{(n-1),\theta}_{j,t+1}(\Omega_{t+1})]$ ($V^{(0),\theta}_H \equiv 0$), where $R_{jt} = \widetilde{R}_{jt} - m_I(x_{jt} + \frac{1}{2}\chi_I x^2_{jt})D^I_{jt}$ with $D^I_{jt} \ge D_{\min} > 0$ from (60).

Step 1 (Stage-game diagonal dominance and attenuated cross-county sensitivity). Since $\mathbf{x}_t$ enters the current dated equilibrium only through $I_t(\ell; \mathbf{x}_t) = \sum_{r=1}^{J} e_{rt}k_{rt}(\ell)X_{rt}$ with $X_{rt} = m_I(x_{rt} + \frac{1}{2}\chi_I x^2_{rt})$, the chain rule gives $\partial^2_{x_j x_j}\widetilde{R}_{jt} = (\partial_{X_{jt}}\widetilde{R}_{jt})m_I\chi_I + (\partial^2_{X_{jt}X_{jt}}\widetilde{R}_{jt})m^2_I(1 + \chi_I x_{jt})^2$. When construction absorbs a small fraction of manufacturing capacity ($m_I = O(\varepsilon)$ with χ_I large so $|\partial_{X_{jt}}\widetilde{R}_{jt}| \le \frac{1}{2}D^I_{jt}$), differentiating R_{jt} twice yields $\partial^2_{x_j x_j}R_{jt} \le -\frac{1}{2}m_I\chi_I D^I_{jt} + O(m^2_I(1 + \chi_I)^2)$ and $\partial^2_{x_i x_j}R_{jt} = O(m^2_I(1 + \chi_I)^2)$ ($i \ne j$). When $H^{(n-1)}_{jt}(\overline{\mathbf{T}}_{t+1}; \mathbf{z}_t, \overline{\mathbf{z}}_t) = \mathbb{E}^{\theta}_t[V^{(n-1),\theta}_{j,t+1}]$ is concave in $\overline{T}_{j,t+1}$ with $|\partial^2_{\overline{T}_i\overline{T}_j}H^{(n-1)}_{jt}| \le C_2\varepsilon^2/(\overline{T}_{i,t+1}\overline{T}_{j,t+1})^{(1+\alpha_T)/\alpha_T}$, the stage-$n$ pseudogradient $g^{(n),\theta}_{jt} = \partial_{x_j}\Pi^{(n),\theta}_{jt}$ is strongly monotone with modulus $\mu = \frac{1}{8}m_I\chi_I D_{\min} > 0$. By Proposition 4, $\mathbf{x}^{(n),\theta}_t$ is unique, and the implicit function theorem on each active interior set $\mathcal{A} = \{r : 0 < x^{(n),\theta}_{rt} < \bar{x}_{rt}\}$ yields the attenuated stock-sensitivity bound (74): $\|\partial_{\overline{T}_{it}}\mathbf{x}^{(n),\theta}_t\| \le C_u\varepsilon/(\mu \min_r \overline{T}^{(1+\alpha_T)/\alpha_T}_{rt})$.

Step 2 (Backward induction on county value concavity). Differentiating $V^{(n),\theta}_{jt}(\Omega_t) = \Pi^{(n),\theta}_{jt}(\mathbf{x}^{(n),\theta}_t(\Omega_t); \Omega_t)$ with respect to $\overline{T}_{jt}$ via the envelope theorem eliminates $\partial_{x_j}\Pi^{(n),\theta}_{jt}\partial_{\overline{T}_j}x^{(n),\theta}_{jt} = 0$ a.e., leaving $\partial_{\overline{T}_j}R_{jt} + \beta\partial_{\overline{T}_j}H^{(n-1)}_{jt}$ plus rival feedback $\sum_{h\ne j}\partial_{x_h}\Pi^{(n),\theta}_{jt}\partial_{\overline{T}_j}x^{(n),\theta}_{ht}$. Because carrier and route-loss savings exhibit diminishing returns $\partial^2_{\overline{T}_j\overline{T}_j}R_{jt} \le -c_R\varepsilon/\overline{T}^{(1+2\alpha_T)/\alpha_T}_{jt} < 0$ and expectations

smooth rival policy kinks against the bounded-score truncated-Gaussian kernel of $\mathbf{z}_t$, (74) restricts the rival feedback derivative to $O((\varepsilon/\mu)\varepsilon^2/(\overline{T}_{it}\overline{T}_{jt})^{(1+\alpha_T)/\alpha_T})$, preserving own-stock concavity and small cross-stock derivatives of $H^{(n)}_{j,t-1}$ for $n = 1, \ldots, H$.

Step 3 (Balanced-growth corridor and infinite-horizon convergence). Rescaling county stocks and increments by $g^t = 1.02^t$ via $\widetilde{T}_{jt} = g^{-t}\overline{T}_{jt}$ and $\bar{d}_j = g^{-t}d_{jt} = |\mathcal{C}_j|^{-1}\int_{\mathcal{C}_j}\iota_{j,1881}(\ell)\,d\ell$ gives $\widetilde{T}_{j,t+1} = g^{-1}(\widetilde{T}_{jt} + \bar{d}_j x_{jt})$. Discounted marginal carrier savings balance $m_I(1+\chi_I x_{jt})D^I_{jt}$ in the interior branch of (61) on a compact corridor $\mathcal{K} = \prod_{j=1}^{J}[L_j, U_j]$ at intensities $x^{(n),\theta}_{jt} \in [\underline{a}_j, \overline{a}_j] \subset (0,1)$ satisfying $(g-1)L_j/\bar{d}_j \le \underline{a}_j \le \overline{a}_j \le (g-1)U_j/\bar{d}_j$, which maps $\mathcal{K}$ into itself. Along $\mathcal{K}$, the state-weight drift is $\kappa = g^{1-\alpha_T}$ and $BL_a = O(\varepsilon/\mu)$, so $\beta\kappa + BL_a = \beta g^{1-\alpha_T} + O(\varepsilon/\mu) < 1$, and $(V^{(H),\theta}_{jt}, \mathbf{x}^{(H),\theta}_t)$ contract geometrically as $H \to \infty$ to a unique infinite-horizon Markov perfect equilibrium. [← Back to Proposition 7] □

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